



Robust and fast estimation of equivalent circuit model from noisy electrochemical impedance spectra

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ABSTRACT

The Equivalent Circuit Model (ECM) is a powerful technique to quantitatively analyze and compare Electrochemical Impedance Spectroscopy (EIS) data. In practice, noise is prevalent in EIS data, due to fuel cell system fluctuations, limited measuring time, and instruments, challenging the accuracy of ECM. There are algorithms that work well on noisy data, yet in many cases, either run time or robustness remains a problem. In this paper, we proposed a robust and fast algorithm that detects outliers, weighs the EIS data, and automatically fits an ECM within a second. For both experimentally measured and simulated noisy EIS data, the new algorithm reduced the impact of noise drastically. The algorithm is demonstrated in Python as part of the open-source software EISART at github.com/leehangyue/EISART.

1. Introduction

1.1. Electrochemical impedance spectroscopy

Electrochemical Impedance Spectroscopy (EIS) is a useful technique in studies on batteries and fuel cells. By injecting alternating excitation current and recording voltage response, EIS collects information on transport processes over a broad range of relaxation time scales that simultaneously happen, such as diffusive mass transport and charge transfer reactions. In a common EIS test, the object under test, for instance, a fuel cell, is assumed to be a linear system, whose response is irrelevant to excitation amplitude. Due to the nature of electrochemistry, however, the response of a fuel cell is inherently nonlinear. As the excitation current amplitude decreases, the nonlinearity of the fuel cell weakens at the cost of a lower Signal-to-Noise Ratio (SNR). Therefore, the excitation current amplitude is commonly set between 0.5% and 2% of the fuel cell nominal operating current to balance the impact of nonlinearity and noise. In fuel cell or electrolyzer systems, stronger noise due to inter-component interaction may be present at certain ranges of frequency, precision electrochemical workstation may be absent and impedance spectra may be measured with less accurate yet faster approaches, such as pseudo random binary sequence [1,2]. Therefore, stronger noise can be inevitable in measured EIS of a practical system.

1.2. Distribution of relaxation time

On a Nyquist plot, relaxation processes with a time scale less than a decade apart may appear as a single semi-circle and are not easily separable. To enhance resolution in time scale, Distribution of Relaxation Time (DRT) was proposed, facilitating the separation of electrode processes with similar time scales, such as the two reaction stages of oxygen reduction on Solid Oxide Fuel Cell (SOFC) cathodes noticed by Osinkin [3], and the 5 different loss mechanisms in full SOFCs observed by Leonide et al. [4].

In general, DRT analysis is based on the integral Eq. (1). $Z(\omega)$ is the complex impedance at angular frequency ω , R_{inf} is the series resistance that characterizes processes with negligible relaxation time (infinite characteristic frequency, hence the name), $\gamma(\tau)$ is the logarithmic distribution density of the relaxation process with relaxation time around τ , which helps distinguish relaxation processes with even closer time scales.

$$Z(\omega) = R_{inf} + \int_{-\infty}^{+\infty} \frac{\gamma(\tau) d\ln\tau}{1 + j\omega\tau} \quad (1)$$

Solving Eq. (1) is an inherently ill-posed mathematic problem that is highly sensitive to noise. To enhance solution stability and reliability, regularization methods are used, among which Tikhonov regularization was proved effective in studies on Solid Oxide Fuel Cells (SOFCs) [5–7].

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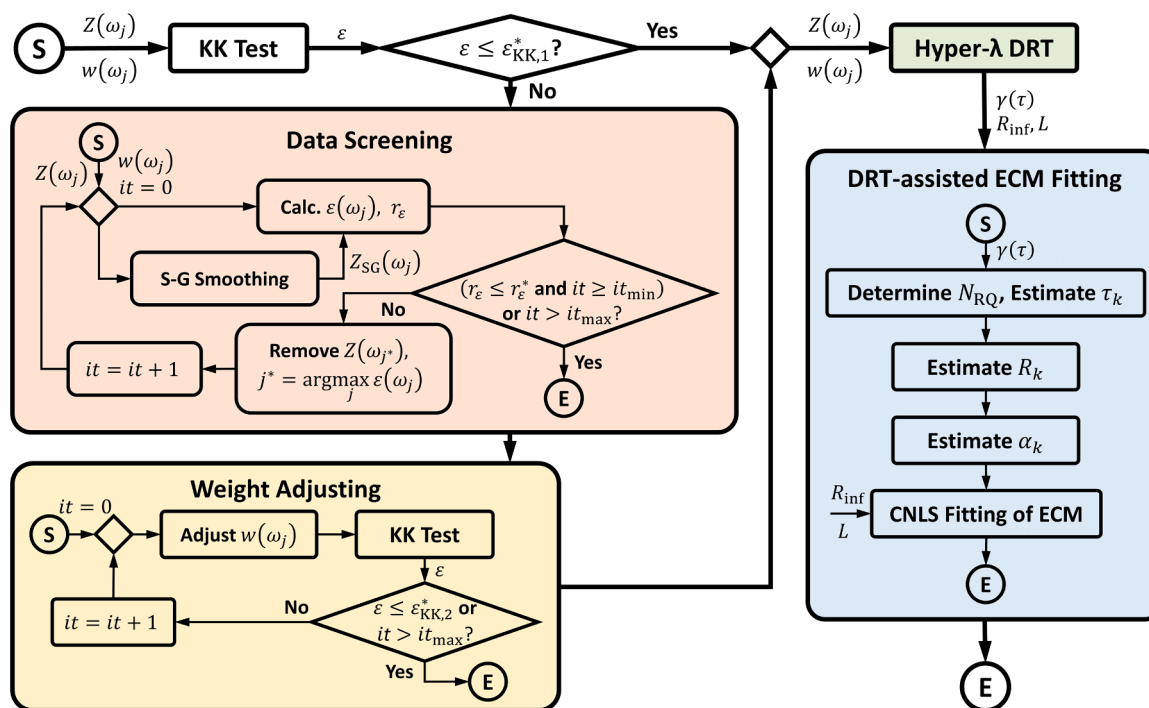


Fig. 1. Flow diagram of the new algorithm. Circled S denotes start and circled E denotes end.

Zhang et al. [8] employed Tikhonov regularization to analyze the imaginary part of impedance spectra and notably enhanced noise resistance of DRT. Wan et al. [9] incorporated Tikhonov regularization and proposed an algorithm based on radial basis function. The solution of Eq. (1) was therefore converted to a finite-dimension optimization problem, enabling a reliable solution of DRT using both real and imaginary parts of impedance spectra. Besides the methodology, Wan et al. [9] studied the choice of the regularization parameter, which blurs resolution in time scale when increased, and weakens robustness to noise when decreased.

As was suggested by Wan et al. [9], a higher noise level requires stronger regularization for minimal error in DRT. However, the quantitative selection method of the regularization parameter remained unresolved. To overcome this, Zhang et al. [10] proposed a method free from the regularization parameter, by including the selection of the regularization parameter as part of the optimization problem. Effat and Ciucci [11] proposed a similar method that automatically adjusts the regularization parameter, namely the Hyper- λ method. Moreover, they presented the Hyper- w method that weighs the input EIS data according to residual. However, the approach by Zhang et al. [10] may result in a mismatch between the total resistance in DRT and EIS, as was pointed out by Gavriluk et al. [12]. Nevertheless, the method by Effat and Ciucci [11] was proved effective on a wide variety of impedance spectra ranging from Solid Oxide Fuel Cell (SOFC) electrodes [13] to double-layer capacitors [14]. Moreover, their method was further augmented with deep-learning [15], Bayesian models [16], and considerations for nonlinearity in the distribution of diffusion time analysis of supercapacitors [17]. Recently, Huang et al. [18] proposed a hierarchical Bayesian approach that further boosted the robustness of DRT against noise, yet this approach may take up to 81.1 s on a common personal computer to estimate DRT, according to the code tutorial on Github mentioned in the research article [18]. Therefore, a robust yet faster approach is still in need for applications on industrial monitoring or batch analysis of impedance spectra.

1.3. Equivalent circuit model

An Equivalent Circuit Model (ECM) is an analytical system that exhibits similar behavior upon perturbations to the real electrochemical device under test. Ideally, the ECM would reproduce the impedance spectra of the test object, so that the equivalent circuit model can predict the dynamic response of the test object and indicate the key parameters of the test object for studies on varied operating conditions or preparation processes. To extract ECM parameters from impedance spectra, Complex Nonlinear Least Squares (CNLS) is employed to automatically determine the ECM parameters that best fit the measurements. In occasions where only one or two relaxation process is prominent, CNLS works well from a random initial guess of model parameters [19] with satisfying accuracy, given that no wild outlier is present in the impedance spectra [20].

For more complicated models, CNLS may end up with different results given different initial guesses, because the objective function that CNLS minimizes has multiple local minima [21]. To increase the chance to obtain physically meaningful ECM parameters, the initial guess can be specified a priori. To fit a large number of correlated impedance spectra, Bondarenko [22] manually specified ECM parameters for the first impedance spectrum as initial guess, run CNLS fit, and use the resulting ECM parameters as the initial guess for the next fit. The Nyquist and Bode plot of an impedance spectrum was also a good graphical indicator of ECM parameters, as was demonstrated by Huang et al. [23] by adjusting the shape and position of impedance spectra of standard ECM elements each at a time. Wang et al. [24] estimated ECM parameters for Li-ion batteries with a predefined model as an initial guess and refined the ECM parameters to obtain the state of charge and temperature of a Li-ion battery.

In the absence of manual interference or a predefined model, DRT was proved useful to fit an ECM. Sumi et al. [25] determined the number of processes and estimated their time constants from corresponding peaks on DRT plots. They identified 5 well-separable processes on a microtubular SOFC, performed CNLS fit for the 5 corresponding equivalent resistances, and obtained physically meaningful results.

When fine-tuning well-defined models with abundant test data,

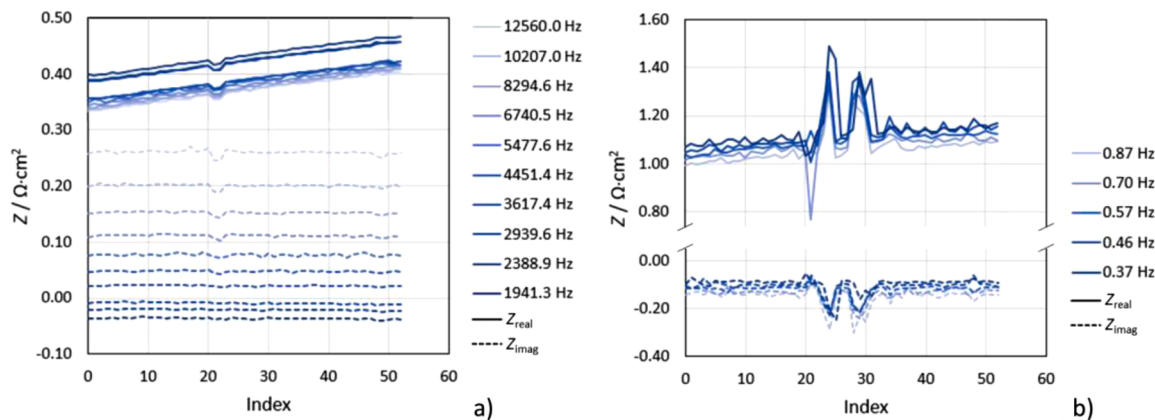


Fig. 2. Trend of real and imaginary parts of impedance at selected frequencies versus measurement index. (a) High frequencies. (b) Low frequencies.

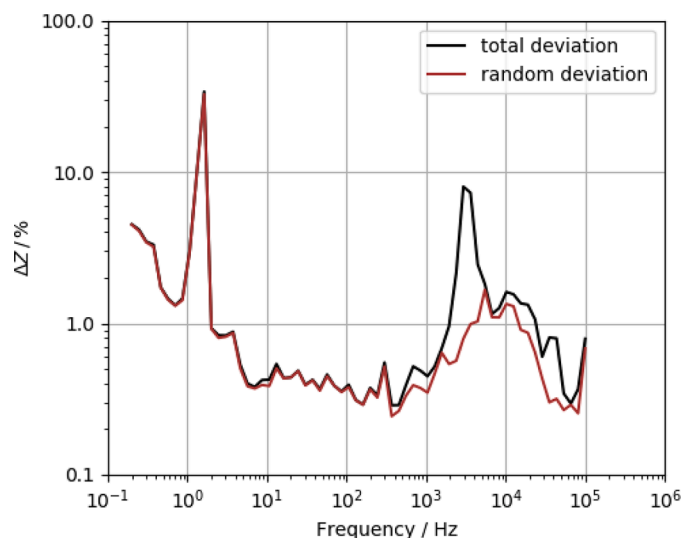


Fig. 3. Estimated noise distribution in the measured impedance spectra versus frequency.

existing models and correlation among test data may effectively ease ECM fitting. Dierickx et al. [26] stated that the corresponding physical process of each peak can be identified with the help of DRT and experiments under varied operating conditions, and the identified physical process can in turn serve as an interpretable starting point of CNLS fits of ECM. In fact, Klotz et al. has proposed a similar method much earlier [27]. In such a fashion, Mertens and Granwehr [28] explored the feasibility of simultaneously analyzing DRT results of a series of synthetic impedance spectra at varied temperatures. The correlation among different impedance spectra effectively reduced the error of ECM parameter estimation. Papurello et al. [29] demonstrated the use of DRT for physiochemically meaningful ECM with measured impedance spectra of a SOFC. In light of DRT-assisted ECM, Papurello et al. [29] went beyond ECM parameters and extracted activation energies of multiple electrode processes.

However, for an unprecedented test object or familiar object tested under unusual conditions, the test object may behave differently from existing models. The number of processes, range of resistance, and relaxation time may need to be refreshed with DRT. Besides, the processes may overlap in a DRT plot, blurring peak positions, hence the initial guess of ECM parameters for CNLS fit. Therefore, a robust algorithm is desired for an initial guess of ECM parameters.

1.4. Our approach

Based on the above-mentioned research, this study demonstrated a robust and fast ECM fitting algorithm facilitated with modified DRT inspired by the work of Liu et al. [16]. Instead of processing the raw impedance data, our approach screens out the outliers before evaluating the DRT, and features improved robustness against individual “wild” data points. The remaining data enter the subsequent KK test to yield frequency-resolved residuals as a measure of data quality, enabling frequency-resolved weight adjustment. After evaluating the DRT and estimating the ECM parameters, the subsequent refinement of ECM parameters further utilizes the weights, minimizing the errors due to noise. With highlights on data screening, KK test-based weight adjustment, and DRT-assisted ECM fitting, the new algorithm also features low computational complexity and high robustness against noise. The functionality of this algorithm has been validated and demonstrated in research articles on Solid Oxide Fuel Cells (SOFCs) using DRT on button cells [30,31] and a large planar cell [32], as well as using ECM on button cells [33] and large planar cells [34,35]. This algorithm was implemented in Python programming language with Graphical User Interface (GUI) and tested with both measured and synthesized impedance spectra.

2. Methodology

Fitting an ECM to an impedance spectrum is a non-linear problem whose numerical solution can be highly sensitive to the initial guess. To solve this problem, a new multi-step algorithm was proposed. Fig. 1 shows the flow diagram of the new algorithm. To evaluate the overall data quality, the weighed KK test is applied to the input impedance spectrum $Z(\omega_j)$ and the corresponding initial weights $w(\omega_j)$. The weights are calculated from the quality index, for example “significance” as in the case of the Zennium electrochemical workstation by Zahner GmbH, or simply initialized as ones. Data points with higher quality indices obtain more weight.

The boxes labeled “KK Test” denote weighed KK test. The weighed KK test in the very beginning analyzes the input impedance spectrum and returns an estimated impedance spectrum $Z_{\text{KK}}(\omega_j)$. After the weighed KK test, the residuals $\varepsilon(\omega_j) = |Z(\omega_j) - Z_{\text{KK}}(\omega_j)|$ are pooled into their Root Mean Squares (RMS) $\varepsilon = \sqrt{\sum_{j=1}^N \varepsilon(\omega_j)^2 / N}$, as an overall quality indicator of the input impedance spectrum. $w(\omega_j)$ is the weight at the j th frequency point. A greater ε corresponds to lower data quality. ε is then compared to a threshold $\varepsilon_{\text{KK},1}^*$. If $\varepsilon \leq \varepsilon_{\text{KK},1}^*$, then the input impedance spectrum and the corresponding weights are considered accurate enough and directly passed on to the DRT analysis. Otherwise, the input impedance spectrum requires Impedance Data Screening and Weight Adjusting before being analyzed with DRT. In this study, $\varepsilon_{\text{KK},1}^* =$

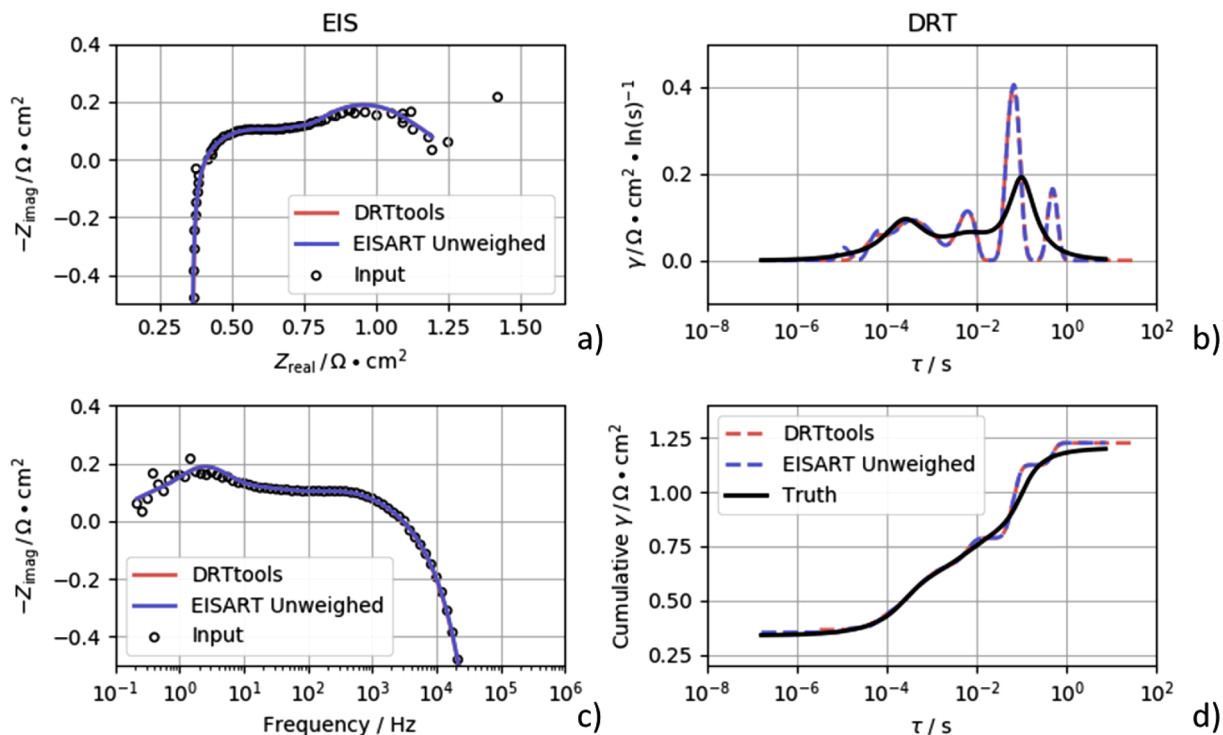


Fig. 4. Comparison of unweighed DRT algorithms in DRTtools and EISART on the 4RQ ECM in section 0. (a) Nyquist plot and (c) half-log Bode plot of the input and regenerated impedance spectra from weighed and unweighed DRT. (b) Distribution function $\gamma(\tau)$ of relaxation time and (d) cumulative distribution function of relaxation time from weighed and unweighed DRT.

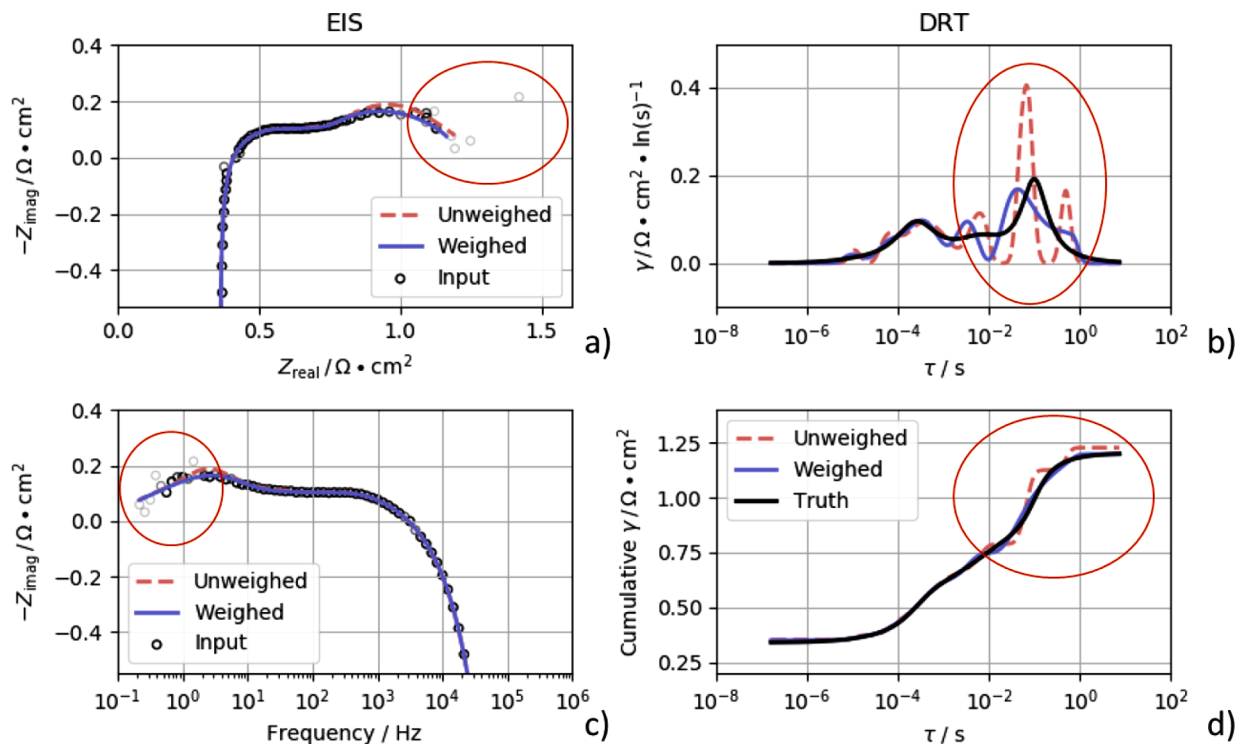


Fig. 5. Comparison of weighed and unweighed DRT algorithms on the 4RQ ECM in section 0. (a) Nyquist plot and (c) half-log Bode plot of the input and regenerated impedance spectra from weighed and unweighed DRT. Lighter color on input data denotes lower weight. (b) Distribution function $\gamma(\tau)$ of relaxation time and (d) cumulative distribution function of relaxation time from weighed and unweighed DRT.

0.2%.

After the DRT analysis, the distribution function $\gamma(\tau)$ of relaxation time τ , along with the series resistance R_{inf} (the subscript “inf” denotes

the real part of the impedance of the ECM at infinite frequency) and series inductance L are passed on to fit the remaining parameters of the ECM, including R_k , τ_k , and α_k of each RQ element (indexed with k). For

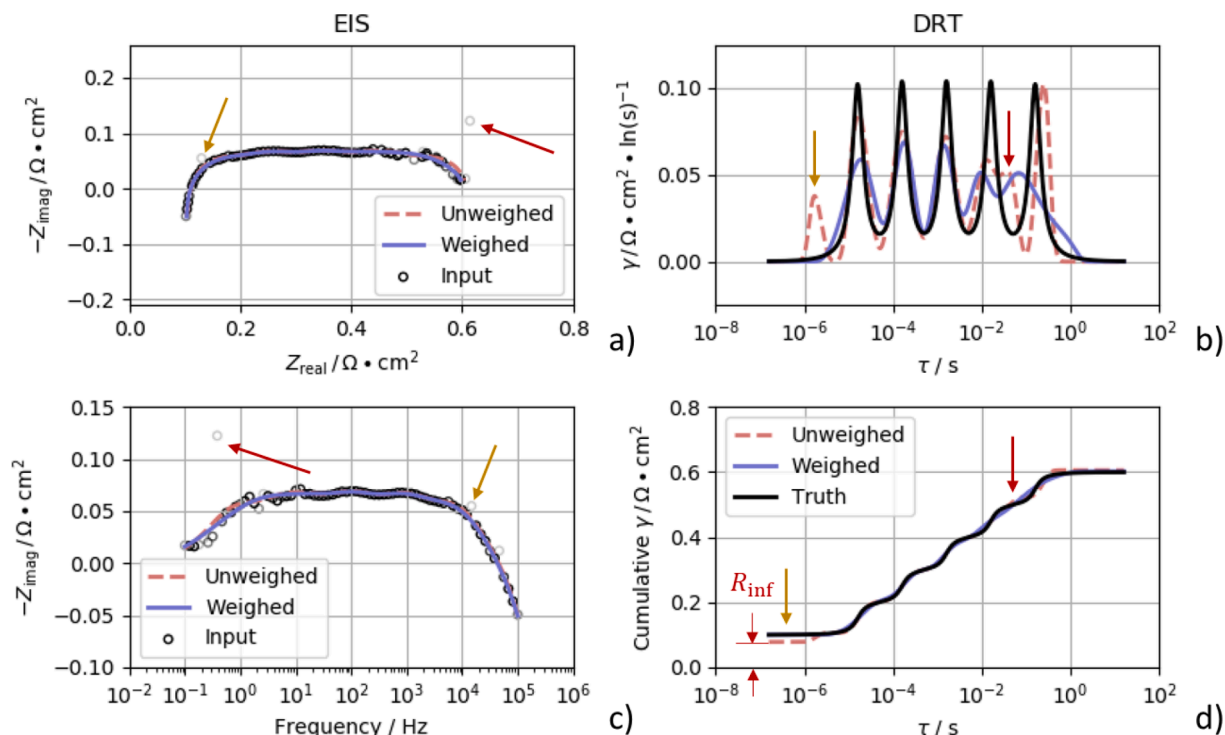


Fig. 6. Comparison of weighed and unweighed DRT algorithms on the 5RQ ECM in section 0. (a) Nyquist plot and (c) half-log Bode plot of the input and regenerated impedance spectra from weighed and unweighed DRT. Lighter color on input data denotes lower weight. (b) Distribution function $\gamma(\tau)$ of relaxation time and (d) cumulative distribution function of relaxation time from weighed and unweighed DRT.

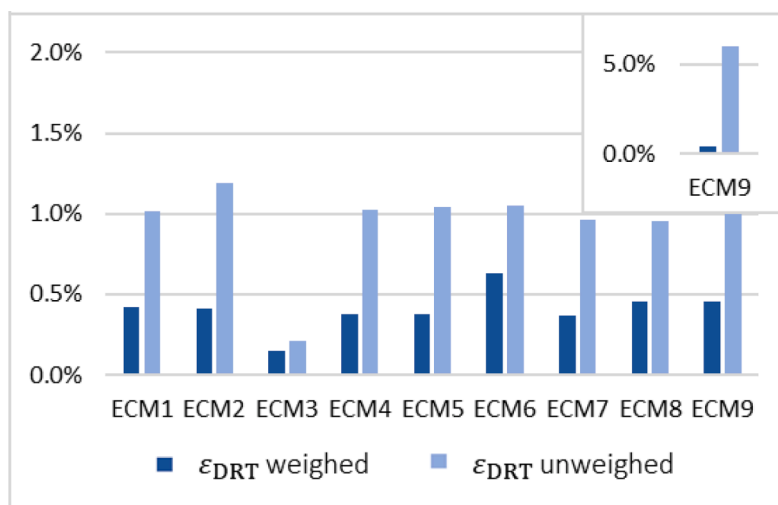


Fig. 7. Average error of regenerated impedance spectra from DRT of the weighed and unweighed algorithm.

detailed description of each step in Fig. 1, please find the following sections of this chapter.

3. Validation: noise modeling

In this chapter, a realistic noise model was developed to generate a massive number of impedance spectra to validate the accuracy of the proposed algorithm. Frequency-dependent noise level was extracted and the occurrence and the standard deviation of “wild” points was modeled based on real measurements.

3.1. Noise measurement

In practice, impedance spectra may contain outlying data points wildly deviating from the rest. Such outliers may result from disturbance at a certain time or frequency, and may considerably corrupt DRT results or lead to difficulty fitting ECMs. To demonstrate the existence of outliers in practice and capture their characteristics, impedance spectra were collected from a $10 \times 10 \text{ cm}^2$ anode-supported SOFC. The SOFC comprised a Ni-YSZ (Yttria-Stabilized Zirconia) anode, a YSZ electrolyte, a GDC (Gadolinium-Doped Ceria) barrier layer, and an LSCF (Lanthanum Strontium Cobalt Ferrite) cathode. Further details about the testbench and the manufacturing process of the fuel cell can be found in an earlier publication [31]. As shown in Fig. 22, 52 consecutive

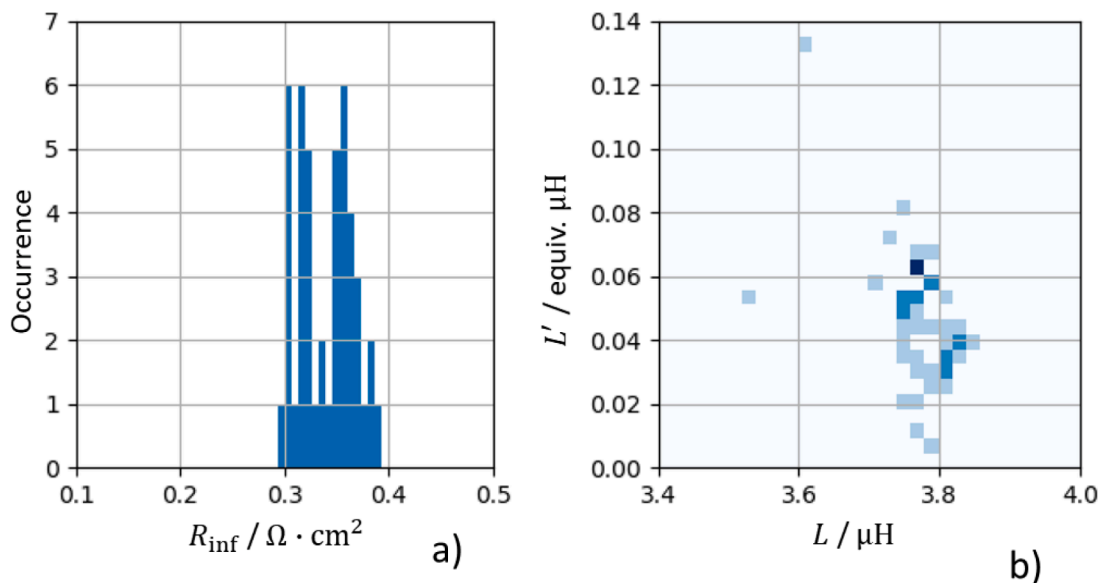


Fig. 8. The distribution of estimated (a) R_{inf} , b) L , and L' of the ECM of the measured impedance spectra. The ECM parameters were determined by the proposed method.

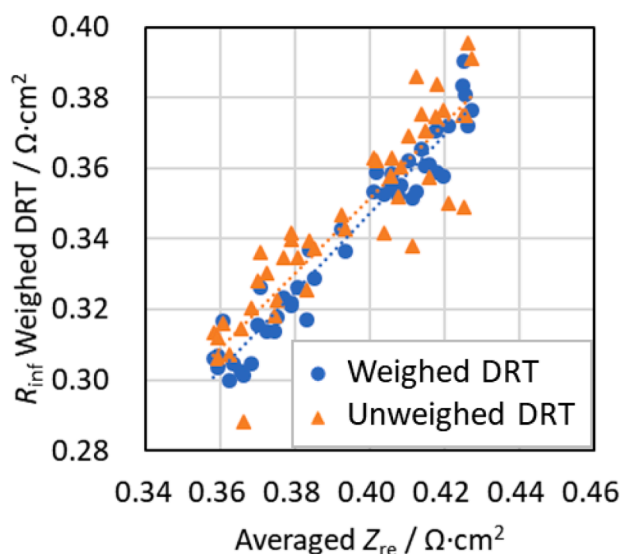


Fig. 9. R_{inf} estimated with weighed DRT and unweighed DRT. The dotted lines are linear regression lines of the corresponding data.

impedance spectra $Z_i(\omega)$, $1 \leq i \leq 52$ were recorded every 15 min. During the measurement, 0.9 Standard Liter per Minute (SLM) dry hydrogen with additional 10% steam and 3.0 SLM dry air was supplied to the SOFC at its Open Circuit Voltage (OCV) in a furnace set at 800°C. The impedance spectra were measured with a Zahner Zennium electrochemical workstation in galvanostatic mode with a single frequency sine perturbation current of 1 Ampere one-sided peak amplitude.

In Fig. 22, the impedances above 1 kHz vary along the real axis and align well along the imaginary axis, while at lower frequencies, especially around 1 Hz, the impedances vary significantly along both the real and the imaginary axis. Fig. 2 shows the trend of impedances at selected frequencies. An increasing trend of the real part is noticeable for all high frequencies in Fig. 2a. Besides the trend in the real part, there was a significant change of impedance at lower frequencies in Fig. 2b for measurements indexed from 20 to 30. The change was later found related to steam content fluctuation due to gas bubbles in the water inlet tube of the steam generator. To estimate the uncertainty of impedance

measurement, impedance spectra indexed 21–24, 27–30 were considered invalid and dismissed. The remaining 44 measurements were analyzed in the following analysis.

The systematic error $e_{\text{sys}}(\omega)$ denotes the error component that remains constant among impedance spectrum samples. To exclude the trend in the measurements, $e_{\text{sys}}(\omega)$ is defined as the averaged residuals of KK test, as shown in Eq. (2).

$$e_{\text{sys}}(\omega) = \text{average}_i (Z_i(\omega_j) - Z_{\text{KK},i}(\omega_j)) \quad (2)$$

To estimate the distribution of random noise at different frequencies despite the trend of the real part, the Weighed KK Test was applied to each impedance spectrum. The relative residuals were calculated with Eq. (3), i is the measurement index, j is the frequency index, $Z_{\text{KK},i}(\omega_j)$ is the KK test transformed impedance spectrum, $\text{average}_i Z_i(\omega_j)$ is the averaged impedance spectrum over all valid measurements, $|\cdot|$ is the modulus operator. The estimated total deviation $\Delta Z(\omega_j)$ is calculated with Eq. (4) as the average relative residuals over all valid measurements, and the random deviation $\sigma_{\Delta Z}(\omega_j)$ is calculated with Eq. (5) to exclude the impact of systematic bias, as shown in Fig. 3. Note that the peak of total deviation between 1 kHz and 10 kHz is significantly higher than random deviation, hence this peak stems largely from systematic bias rather than random disturbance.

$$\Delta Z_i(\omega_j) = \frac{Z_i(\omega_j) - Z_{\text{KK},i}(\omega_j)}{|\text{average}_i Z_i(\omega_j)|} \quad (3)$$

$$\Delta Z(\omega_j) = \sqrt{\text{average}_i |\Delta Z_i(\omega_j)|^2} \quad (4)$$

$$\sigma_{\Delta Z}(\omega_j) = \sqrt{\text{average}_i \left| \Delta Z_i(\omega_j) - \text{average}_i \Delta Z_i(\omega_j) \right|^2} \quad (5)$$

3.2. Noise simulation

The noise in measured impedance spectra comprises frequency-dependent random noise and systematic bias that only affects certain frequencies. To validate the new algorithm, ground truth impedance spectra Z_{GT} were generated from ECM with known parameters according to Eq. (47). The ground truth ECM comprises R_{inf} representing ohmic

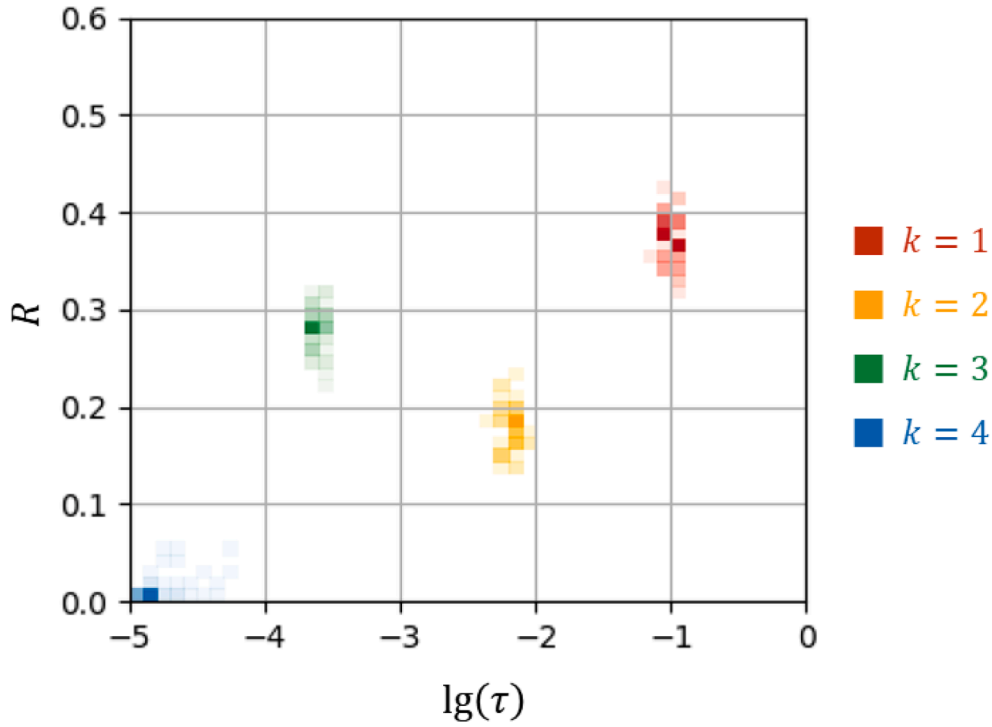


Fig. 10. The distribution of estimated R_k and τ_k of each RQ element in the ECM of the measured impedance spectra.

Table 1

Parameters of the 4RQ reference ECM.

RQ Element Index k	$R_k / \Omega \cdot \text{cm}^2$	τ_k / ms	α_k
1	0.39	100	0.79
2	0.18	6.4	0.65
3	0.28	0.25	0.70
4	0.015	0.015	0.63
	$R_{\text{inf}} / \Omega \cdot \text{cm}^2$	$L / \mu\text{H}$	$L' / \text{equiv. } \mu\text{H}$
	0.34	3.76	0.0515

resistance, L representing parasitic inductance, and RQ (also named ZARC) elements featuring direct-current resistance R_k , characteristic time constant τ_k , and phase exponent α_k . Systematic bias and proportional random noise were added to the ground truth according to Eq. (6).

$$Z_{\text{syn}}(\omega) = Z_{\text{GT}}(\omega) + |Z_{\text{GT}}(\omega)| \cdot e_{\text{rand}}(\omega) + e_{\text{sys}}(\omega) \quad (6)$$

The random noise $e_{\text{rand}}(\omega)$ comprises broadband noise $e_{\text{rand,b}}(\omega)$ and narrowband noise $e_{\text{rand,n}}(\omega)$. The broadband noise $e_{\text{rand,b}}(\omega)$ is defined by Eq. (7), ξ_1 and ξ_2 are real-valued independent standard gaussian random variables with zero mean and unit variation, $e_{\text{rand,env}}(\omega)$ is the envelope function of frequency-dependent noise intensity. The narrowband noise $e_{\text{rand,n}}(\omega)$ is applied to randomly chosen frequencies in a certain range. For each chosen frequency ω^* , $e_{\text{rand,n}}(\omega^*)$ is defined by Eq. (8).

$$e_{\text{rand,b}}(\omega) = (\xi_1 + j\xi_2)e_{\text{rand,env}}(\omega) \quad (7)$$

$$e_{\text{rand,n}}(\omega^*) = (\xi_1 + j\xi_2)e_{\text{rand,n,amp}}(\omega^*) \quad (8)$$

4. Results and discussion

4.1. Robust weighed DRT

The DRT calculated with the proposed algorithm was applied to simulated noisy impedance spectra. To investigate the accuracy of DRT, impedance spectra were regenerated from DRT results according to Eq. (9). For reference, the unweighed DRT calculated with the well-known

DRTtools in MATLAB (2020) [9] was compared to the DRT calculated with our software EISART, and plotted in Fig. 4. Despite negligible differences in DRT around $\tau = 10^5$ s, and the extended range of τ in DRTtools, the results from DRTtools and EISART are almost the same. This justifies the conformity of algorithm realization and reproducibility of calculation results of unweighed DRT.

To reveal the effect of weighing, the unweighed version of the proposed algorithm with $w(\omega_j) \equiv 1$ was also applied to the same set of simulated impedance spectra. The ground truth of DRT is calculated with Eq. (51). The results on the 4RQ ECM in section 0 was plotted in Fig. 5. In Fig. 5a and c, the “wild” input points in the red circle were light-colored, implying that they were screened out. Near the circled region, there is a noticeable difference between the regenerated impedance spectrum of unweighed DRT and the input, while the regenerated impedance spectrum of weighed DRT appears closer to the collective trend of the input. In the circled region in Fig. 5b, the unweighed DRT resulted in an oscillating γ that deviates from the ground truth, implying that the unweighed DRT analysis was significantly affected by the “wild” points. In comparison, the weighed DRT analysis effectively reduced the error of estimated γ from the ground truth. In Fig. 5d, the cumulative γ from weighed DRT is significantly closer to the ground truth than from the unweighed DRT. This implies that weighed DRT effectively improves the accuracy of the subsequent initial estimation of R_k of ECM.

The results on the 5RQ ECM, specifically ECM1 in section 0 was plotted in Fig. 6. Since ECM1 has 5 RQ elements with identical parameters, it is convenient to compare the potential impact of DRT on the subsequent ECM fitting of parameters of each RQ element. The “wild” point indicated by the red arrow in Fig. 6a and c resulted in an extra peak in Fig. 6b, disturbing the estimation of n , the number of RQ elements in the ECM. Moreover, the “wild” point indicated by the dark yellow arrow in Fig. 6a and c also resulted in an extra peak in Fig. 6b, leading to an extra RQ element and underestimated R_{inf} . Therefore, the weighed algorithm also effectively improves the estimation accuracy of R_{inf} .

To quantitatively assess the accuracy of DRT, the weighed average error of regenerated impedance spectra relative to the ground truth $Z_{\text{GT}}(\omega_j)$ were calculated according to Eqs. (10) and (11).

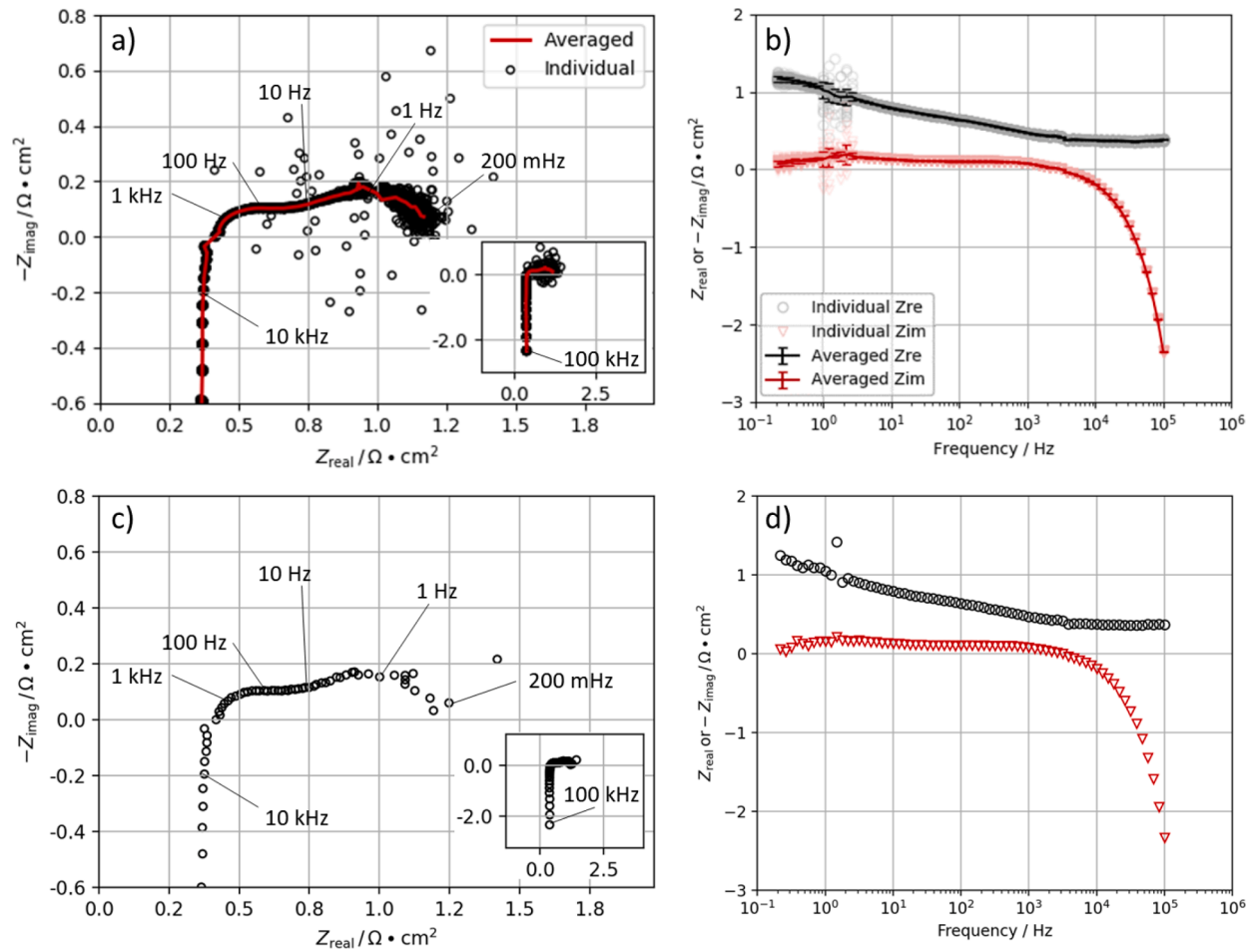


Fig. 11. Simulated impedance spectra of the reference ECM based on the measurements. (a) Nyquist plot of the impedance spectra and their average. (b) The average and standard deviation of the impedance. (c) Nyquist plot of a sample of the impedance spectra. (d) The real and imaginary parts of the sample impedance spectrum.

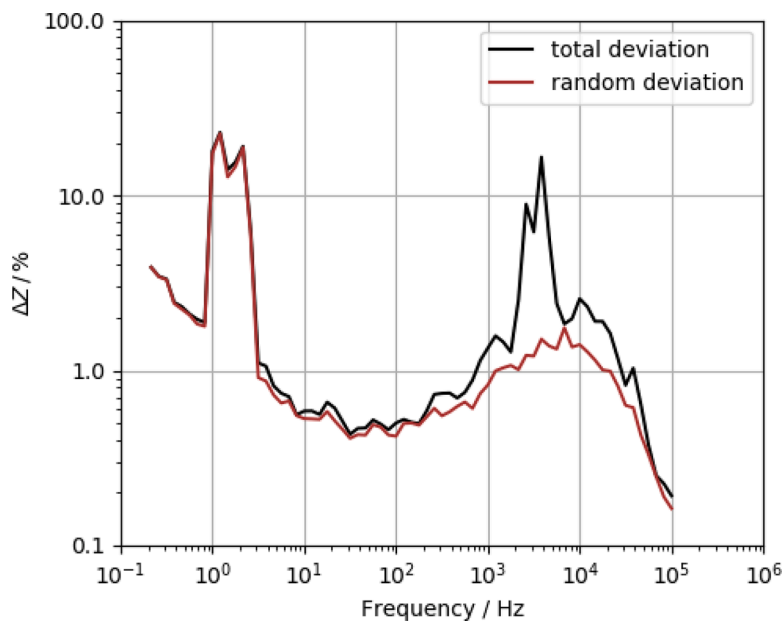


Fig. 12. Noise distribution in the simulated 4RQ impedance spectra versus frequency.

$$Z_{\text{DRT}}(\omega) = R_{\text{inf}} + \int_{-\infty}^{+\infty} \frac{\gamma(\tau) d\ln\tau}{1 + j\omega\tau}$$

$$(9) \quad \varepsilon_{\text{DRT}}(\omega_j) = |Z_{\text{GT}}(\omega_j) - Z_{\text{DRT}}(\omega_j)|$$

$$(10)$$

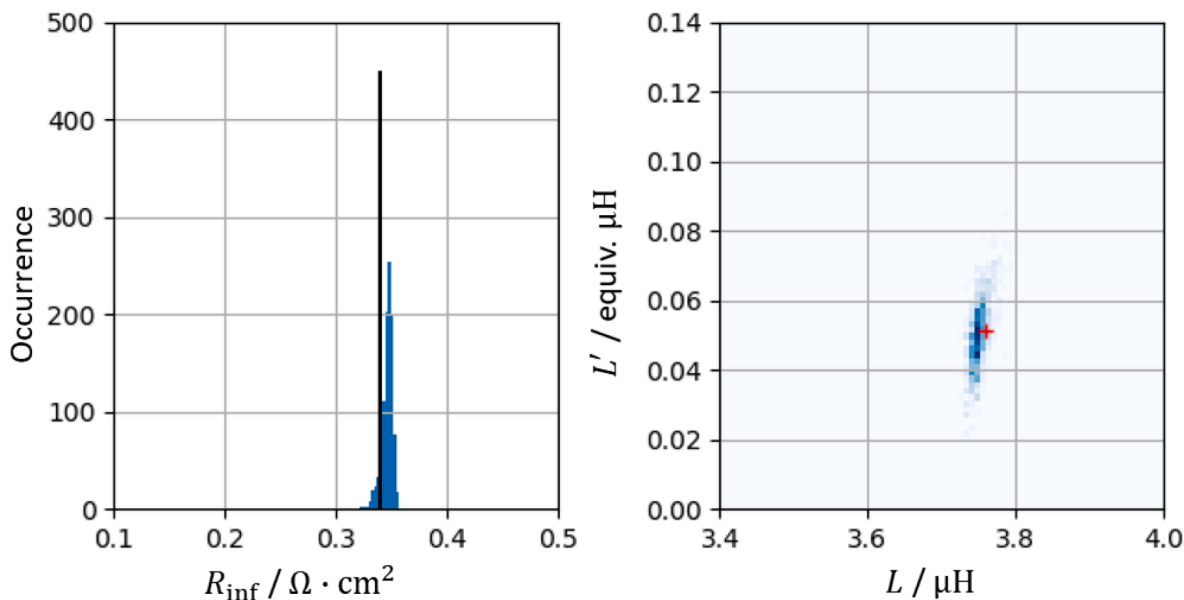


Fig. 13. The distribution of estimated (a) R_{inf} , (b) L , and L' of the ECM of the simulated impedance spectra. The vertical line in (a) and the red “+” sign mark the ground truth.

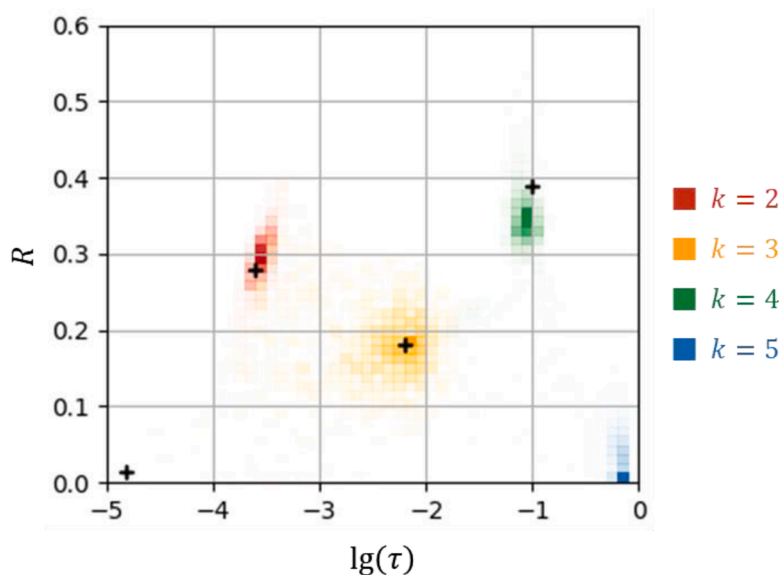


Fig. 14. The distribution of estimated R_k and τ_k of each RQ element in the ECM of the simulated 4RQ impedance spectra. The black “+” signs mark the ground truth.

$$\varepsilon_{\text{DRT}} = \sqrt{\frac{\sum_{j=1}^N (w(\omega_j) \cdot \varepsilon_{\text{DRT}}(\omega_j))^2}{\sum_{j=1}^N (w(\omega_j))^2}} \quad (11)$$

The proposed algorithm was compared to its unweighed version. After applying the proposed method and its unweighed version to the 5RQ ECMs in section 0, the resulting average error ε_{DRT} were plotted in Fig. 7. In all the 9 tested ECMs, the weighed algorithm yields smaller error, implying that weighing improves DRT reliability. This further confirmed that the weighed algorithm has notably improved the accuracy of DRT and hence the reliability of the subsequent ECM fitting.

4.2. Fitting of 4RQ ECM based on measured impedance

The proposed method was applied to the measured impedance spectra in section 0 3.1 Noise Measurement. The results were shown in Fig. 8. (R_{inf} , L , and L') and Fig. 10 (R_k and τ_k of the RQ elements). In

Fig. 8a, the distribution of R_{inf} spread in the range from $0.3 \Omega \cdot \text{cm}^2$ to $0.4 \Omega \cdot \text{cm}^2$. Fig. 9 visualized the correlation of estimated R_{inf} to the real part of the averaged impedances at the frequencies shown in Fig. 2. Recall that the mid-to-high frequency impedances in Fig. 2 display a clear increasing trend with time. From Figs. 2 and 9, it is clear that the spread of ohmic resistance R_{inf} was mainly due to the drift of test object property instead of the uncertainty of R_{inf} estimation. Moreover, in Fig. 9, the R_{inf} estimated with weighed DRT exhibit notably less variance around its regression line, compared to unweighed DRT. This is coherent with section 0 that the weights improve the accuracy of DRT analysis.

Fig. 8b shows the distribution of L and L' , and Fig. 10 shows the distribution of R_k and τ_k of the RQ elements. Since that no obvious correlation was noticed between L and L' , and between R_k and τ_k , the average of each parameter was used to determine a 4RQ reference ECM of the test object. The averages of R_{inf} , L , L' , R_k , τ_k , and α_k were listed in Table 1, in which the averages of τ_k are geometric averages, while the

Table 2

Estimated ECM parameters from the simulated 4RQ impedance spectra.

Parameter	Reference	Average	Standard Deviation
ε_{ECM}	0%	0.88%	0.30%
$R_{\text{inf}} / \Omega \cdot \text{cm}^2$	0.340	0.347	0.005
$L / \mu\text{H}$	3.76	3.75	0.01
$L' / \text{equiv.} \mu\text{H}$	0.0515	0.05	0.01
$R_1 / \Omega \cdot \text{cm}^2$	0.015	-	-
τ_1 / ms	0.015	-	-
α_1	0.63	-	-
$R_2 / \Omega \cdot \text{cm}^2$	0.280	0.275	0.069
τ_2 / ms	0.250	0.285	0.20 dec.
α_2	0.70	0.66	0.06
$R_3 / \Omega \cdot \text{cm}^2$	0.180	0.182	0.045
τ_3 / ms	6.40	3.94	0.43 dec.
α_3	0.65	0.60	0.07
$R_4 / \Omega \cdot \text{cm}^2$	0.390	0.328	0.059
τ_4 / ms	100	67.8	0.38 dec.
α_4	0.79	0.82	0.09
$R_5 / \Omega \cdot \text{cm}^2$	-	0.06	0.10
τ_5 / ms	-	530	0.25 dec.
α_5	-	0.75	0.08

average of R_k and α_k are arithmetic averages. By adding noise to the 4RQ reference ECM, the accuracy of the algorithm in this study was demonstrated.

Based on the noise simulation method in section 0, $N_{\text{sim}} = 1000$ impedance spectra were generated and only 50 spectra were plotted in Fig. 11 for clarity. To simulate the measured noise in Fig. 3, the broadband noise envelope $e_{\text{rand,env}}(\omega)$ is specified by Eq. (12). The narrowband noise was applied to one randomly chosen frequency ω^* satisfying $1 \text{ Hz} \leq \omega^*/2\pi \leq 3 \text{ Hz}$, $e_{\text{rand,n,amp}}(\omega^*) = 2.5 \times 10^{-1}$.

$$e_{\text{rand,env}}(\omega) = 2 \times 10^{-3} + \frac{6 \times 10^{-2}}{1 + \frac{\omega}{2\pi(200 \text{ mHz})}} + \frac{4 \times 10^{-4}}{\left(1 + \frac{2\pi(3 \text{ kHz})}{\omega}\right) \left(1 + \frac{\omega}{2\pi(30 \text{ kHz})}\right)} \quad (12)$$

Compared to Fig. 22a, the simulated impedance spectra in Fig. 11a display no trend of the real part of the impedance, since the 4RQ reference ECM is does not change with time. The uncertainty of the simulated 4RQ impedance spectra was shown in Fig. 11b. While the uncertainty of impedances at most frequencies in Fig. 11b are similar to the measurements in Fig. 22b, the uncertainty between 1 Hz and 3 Hz is lower, except for 1.6 Hz, since the “wild” point at 1.6 Hz in the

measurements were replaced by a randomly chosen point between 1 Hz and 3 Hz. The random selection of “wild” point frequency accounts for the noise distribution while demonstrating the “wild” point detection capability of the new algorithm. For comparison to the measured noise distribution in Fig. 3, the noise distribution of the simulated 4RQ impedance spectra was plotted in Fig. 12. The peak between 1 kHz and 10 kHz is consistent with the measured noise, while the peak between 1 Hz and 3 Hz slightly differs due to the random selection of the “wild” point.

The proposed algorithm was applied to $N_{\text{sim}} = 1000$ simulated 4RQ impedance spectra, the resulting distributions of ECM parameters were visualized in Figs. 13 and 14. Define fitting error ε_{ECM} in Eq. (13) as the average relative difference between the impedance spectrum of the estimated ECM and the ground truth (reference) ECM. Quantitative results were listed in Table 2. Note that the averages and standard deviations of characteristic time τ_k were calculated logarithmically, as specified in Eqs. (14) and (15). The standard deviation of τ_k were given in decades (dec.).

$$\varepsilon_{\text{ECM}} = \sqrt{\frac{\sum_{j=1}^N |Z_{\text{ECM,est}}(\omega_j) - Z_{\text{ECM,ref}}(\omega_j)|^2}{\sum_{j=1}^N |Z_{\text{ECM,ref}}(\omega_j)|^2}} \quad (13)$$

$$\bar{\tau}_k = \exp\left(\frac{\text{average}(\ln(\tau_{k,i}))}{i}\right) \quad (14)$$

$$\sigma_{\tau_k} = \sqrt{\ln^2\left(\frac{\tau_{k,i}}{\bar{\tau}_k}\right)} \quad (15)$$

As shown in Fig. 13, R_{inf} , L , and L' were effectively extracted from the simulated noisy 4RQ impedance spectra with useable precision. In Fig. 14, the RQ element parameters were also recovered, except that the RQ element $(RQ)_1$ was totally neglected, while an extra RQ element $(RQ)_5$ appeared with relaxation time close to 1 s. Nevertheless, the other 3 RQ elements $(RQ)_2$, $(RQ)_3$, and $(RQ)_4$ align well with the ground truth. The average and standard deviations of the estimated ECM parameters were listed in Table 2.

According to Table 2, the estimated R_{inf} displays a small standard deviation of 1.5% of the ground truth and an average only 2% greater than the ground truth. The averages of estimated R_2 and R_3 were also less than 2% away from the ground truths, with standard deviations less than 3% of the respective ground truths. Notably, the average of

Table 3

Parameters of the 5RQ reference ECMs. ECM1 is used for the control group. The parameter in ECM2 through ECM9 that is different from ECM1 is highlighted in bold.

ECM index	1	2	3	4	5	6	7	8	9
R_{inf}/Ω	0.1	0.1	0.1	0.1	0.3	0.1	0.1	0.1	0.1
$L/\mu\text{H}$	0.1	1.0	0.1	0.1	0.1	0.1	0.1	0.1	0.1
R_1/Ω	0.1	0.1	0.1	0.1	0.1	0.3	0.1	0.1	0.1
$\frac{1}{2\pi\tau_1}/\text{Hz}$	10^4	10^4	10^4	10^4	10^4	10^4	10^4	10^4	10^4
R_2/Ω	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
$\frac{1}{2\pi\tau_2}/\text{Hz}$	10^3	10^3	10^3	10^3	10^3	10^3	10^3	10^3	10^3
R_3/Ω	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
$\frac{1}{2\pi\tau_3}/\text{Hz}$	10^2	10^2	10^2	10^2	10^2	10^2	10^2	10^2	10^2
R_4/Ω	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
$\frac{1}{2\pi\tau_4}/\text{Hz}$	10	10	10	10	10	10	30	10	10
R_5/Ω	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
$\frac{1}{2\pi\tau_5}/\text{Hz}$	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
α_k	0.9	0.9	0.9	0.8	0.9	0.9	0.9	0.9	0.9
e_{broad}	1.0%	1.0%	0.1%	1.0%	1.0%	1.0%	1.0%	1.0%	1.0%
e_{narrow}	10%	10%	1.0%	10%	10%	10%	10%	10%	10%
$\frac{\omega_{\text{min}}}{2\pi}/\text{Hz}$	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	1.0
$\frac{\omega_{\text{max}}}{2\pi}/\text{Hz}$	10^5	10^5	10^5	10^5	10^5	10^5	10^5	10^4	10^5

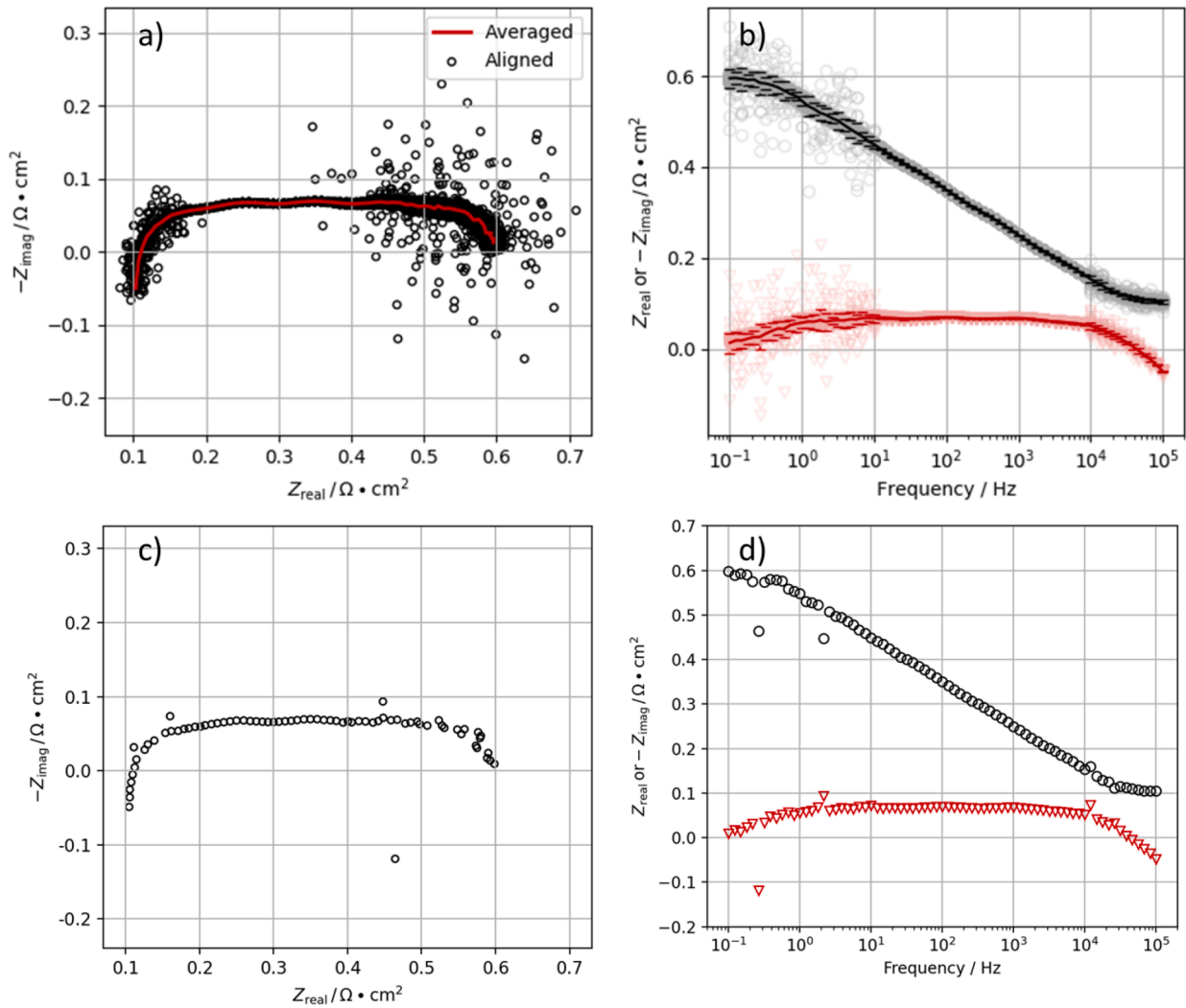


Fig. 15. Simulated impedance spectra of ECM1. (a) Nyquist plot of the impedance spectra and their average. (b) The average and standard deviation of the impedance. (c) Nyquist plot of a sample of the impedance spectra. (d) The real and imaginary parts of the sample impedance spectrum.

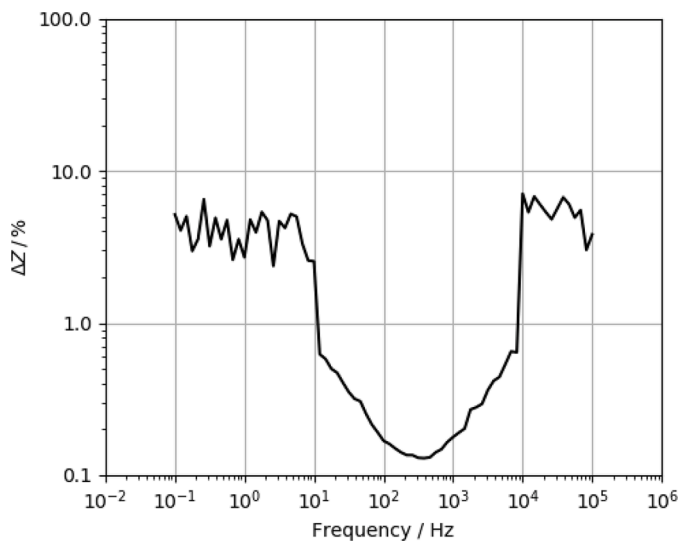


Fig. 16. Noise distribution in the simulated 5RQ impedance spectra of ECM1 versus frequency.

Table 4

Relative average bias ε and standard deviation σ of ECM parameters estimated from ECM1 of the 5RQ ECMs. $\varepsilon \cdot L'$ was calculated relative to L .

Parameter	ε	σ	Parameter	ε	σ
R_{inf}	2.2%	4.9%	R_3	0.5%	5.2%
L	-3.1%	7.7%	$\lg \tau_3$	-0.033	0.200
L'	0.3%	4.8%	α_3	-3.6%	4.3%
R_1	-7.3%	16.6%	R_4	2.4%	7.4%
$\lg \tau_1$	0.018	0.166	$\lg \tau_4$	-0.029	0.230
α_1	2.8%	5.8%	α_4	-4.8%	6.0%
R_2	-1.2%	5.6%	R_5	1.6%	20.8%
$\lg \tau_2$	-0.011	0.159	$\lg \tau_5$	-0.015	0.112
α_2	-2.1%	5.2%	α_5	-3.1%	6.5%

estimated R_4 is significantly smaller than the ground truth. Nevertheless, the sum of estimated $R_4 + R_5 = 0.388 \Omega \cdot \text{cm}^2$ is very close to the ground truth $R_4 = 0.390 \Omega \cdot \text{cm}^2$. This implies that the ground truth R_4 was mistakenly assigned to 2 separate RQ elements. Compared to R_k , the relaxation time τ_k of the RQ elements were estimated less accurately, with the estimated averages farther away from the ground truth and greater standard deviations from 0.20 to 0.43 decades. The spread exponents α_k were estimated with averages and standard deviations around 10% of the respective ground truths.

From the above results, it is concluded that the proposed ECM fitting



Fig. 17. Estimation errors of R_{inf} , L , and L' of the 5RQ ECMs. (a) average biases, (b) standard deviations.

algorithm successfully estimated the underlying ECM parameters. R_{inf} , L , L' , and most R_k were estimated with promising accuracy, while $(RQ)_1$ was neglected, $(RQ)_4$ and $(RQ)_5$ were mistakenly separated. Nevertheless, the accuracy of the proposed algorithm is satisfying, considering the low DC resistance of the neglected $(RQ)_1$ and the surplus $(RQ)_5$. The relation of estimation errors to ECM ground truth and the noise will be discussed in section 0.

4.3. Fitting of 5RQ ECMs with rounded parameters

To investigate the relation of estimation errors to ECM ground truth and the noise, numerical experiments were carried out on simulated noisy impedance spectra of 9 ECMs, with parameters listed in Table 3. The impedance spectra were simulated by adding proportional noise to the respective ground truth impedance spectra evaluated with Eq. (47). The proportional noise were added according to Eqs. (6) through (8). The broadband noise envelope $e_{\text{rand,env}}(\omega)$ is specified by Eq. (12), the systematic error $e_{\text{sys}}(\omega)$ was set to zero. The narrowband noise was applied to 2 randomly chosen frequencies ω^* satisfying $\omega^*/2\pi \leq 10$ Hz and 2 randomly chosen frequencies ω^* satisfying $\omega^*/2\pi \geq 10$ kHz, $e_{\text{rand,n,amp}}(\omega^*) = e_{\text{narrow}}$. The proposed algorithm was tested on each ECM

with $N_{\text{sim}} = 10,000$ repetitions.

$$e_{\text{rand,env}}(\omega) = 1 \times 10^{-3} + e_{\text{broad}} \left(\frac{1}{1 + \frac{\omega}{2\pi(10 \text{ Hz})}} + \frac{1}{\left(1 + \frac{2\pi(10 \text{ kHz})}{\omega}\right)} \right) \quad (16)$$

As an example, 50 simulated impedance spectra of ECM1 and their average were plotted in Fig. 15a. The corresponding standard deviation at each frequency was plotted in Fig. 15b. The noise distribution of the impedance spectra of ECM1 was plotted in Fig. 16. Recall the parameters of ECM1 in Table 3, the relaxation time of $(RQ)_5$ corresponds to 1 Hz and is directly affected by the “wild” points. The relaxation time of $(RQ)_1$ and $(RQ)_4$ correspond to 10 kHz and 10 Hz and are on the edge of the frequency range of “wild” points. The relaxation time of $(RQ)_2$ and $(RQ)_3$ correspond to 1 kHz and 100 Hz and are only affected by mild broadband noise.

The average biases ε and standard deviations σ of all ECM parameters estimated from ECM1 with the proposed algorithm were listed in Table 4. The average biases ε and standard deviations σ of R_{inf} , L , and L' from ECM1 through ECM9 were plotted in Fig. 17. Note that since the ground truth of L' is 0, the relative error of L' was calculated relative to L instead. In Fig. 17a, except $\varepsilon-R_{\text{inf}}$ of ECM1, ECM6, and ECM7, the

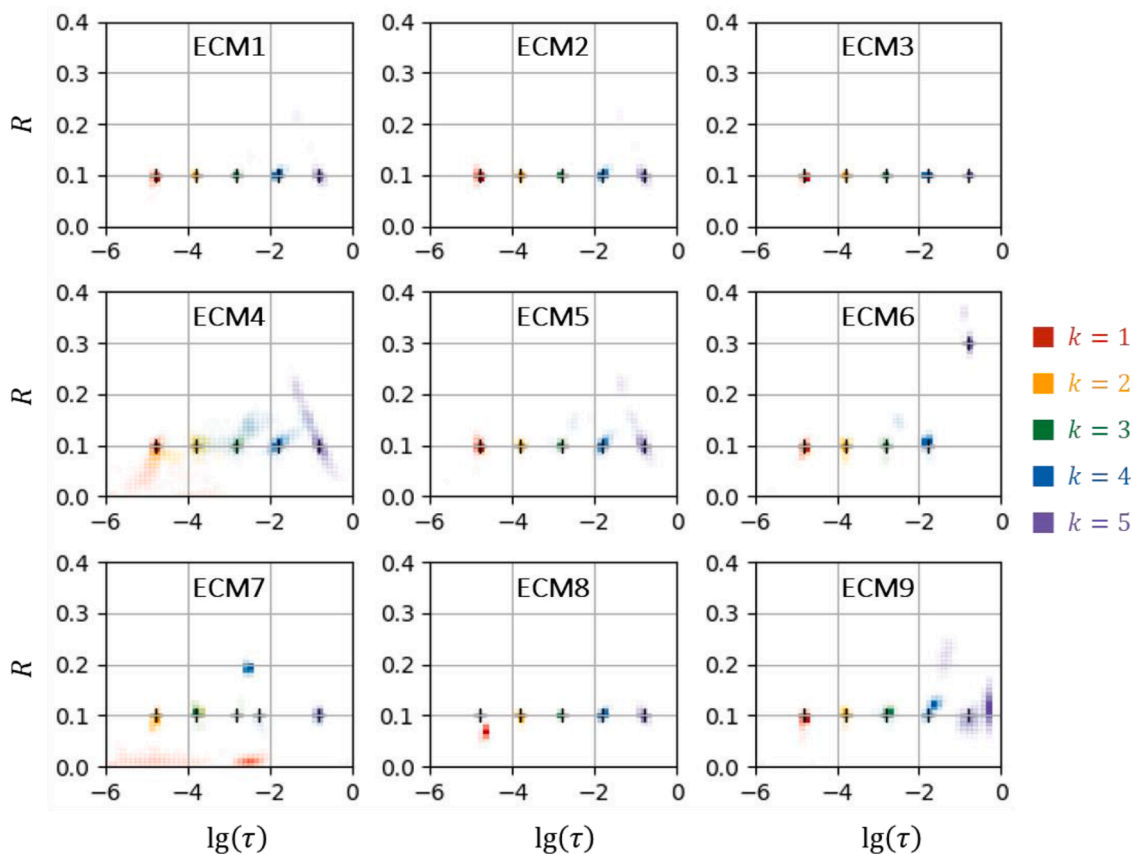


Fig. 18. Distribution of estimated R_k and τ_k of the SRQ ECMs. The black “+” signs mark the ground truth.

average biases of the weighed algorithm were notably smaller or at least equal to the unweighed algorithm. In Fig. 17b, except $\sigma-R_{\text{inf}}$ and $\sigma-L$ of ECM8, the standard deviations of the weighed algorithm were also smaller than the unweighed algorithm. Compared to ECM1, ECM2 features greater inductance, and the average biases of the weighed algorithm were significantly smaller. This implies that pure inductance did not disrupt the estimation of R_{inf} , L , and L' , and may improve the accuracy of estimated R_{inf} .

ECM3 features lower noise level, yet the average biases were greater than ECM1, for both the weighed and unweighed algorithms. This implies that the error may come from algorithm design rather than noise-related factors. Note that both greater R_{inf} and greater and positive L move the high-frequency intersection (intersection of the impedance plotline with the real axis) to the right in a Nyquist plot. Considering that R_{inf} was overestimated while L was underestimated for ECM3 of both weighed and unweighed algorithms, their respective effect on the high-frequency intersection counteract each other. In the absence of impedances at higher frequencies, the impedance can be insensitive to simultaneously increased R_{inf} and decreased L , hence resulting in the biased estimations.

ECM4 features lower α_k than ECM1, the peaks in the DRT of the RQ elements spread wider and thus more difficult to separate. From ECM1 to ECM4, the average biases and standard deviation of R_{inf} and L of the weighed algorithm increased less than the unweighed algorithm, implying that lower α_k can lead to more difficulty fitting an ECM, while weighing alleviated the impact of lower α_k .

ECM5 features greater R_{inf} than ECM1, hence greater impedance magnitude and higher absolute noise amplitude. As a result, the average bias of L' and the standard deviations of L and L' were greater in ECM5. On the contrary, the average biases of R_{inf} and L and the standard deviation of R_{inf} were smaller in ECM5. Consider the greater R_{inf} in ECM5 than ECM1, the decrease of relative average bias and standard deviation

can be attributed to the change in R_{inf} rather than the decrease of absolute error in R_{inf} . Note that the average bias of L decreased while the standard deviation increased, it can be concluded that the noise contributed to the uncertainty of the estimation of L , and the smaller average bias of L can be attributed to the increased chance of hitting the ground truth given greater standard deviation and the vast number of repetitions.

ECM6 features greater R_5 than ECM1, ECM7 features τ_4 closer to τ_3 , ECM9 features a truncated decade at the low-end of frequency. The average biases and standard deviations of R_{inf} , L , and L' displayed no significant change, implying that the estimation accuracy of R_{inf} , L , and L' may not be sensitive to small changes in parameter changes in RQ elements or truncated low frequencies.

ECM8 features a truncated decade at the high-end of frequency. The average biases and the standard deviations of R_{inf} , L , and L' increased drastically. Consider that high frequency impedance is highly sensitive to inductance and is indispensable to the separation of R_{inf} from the RQ elements, the loss of high frequency impedance in ECM8 leads to pronounced difficulty estimating R_{inf} , L , and L' , thus the great estimation error.

From the above analysis, the estimation error of R_{inf} , L , and L' is sensitive to the frequency range at its high-end, noise level, and the ground truth value of R_{inf} and L . Moreover, the weighed algorithm outperformed the unweighed algorithm significantly. To be succinct and highlight the influencing factors on estimating RQ element parameters, only the weighed algorithm was demonstrated in the following discussion.

The RQ element parameters estimated with the weighed algorithm were plotted in Fig. 18 for R_k , τ_k and Fig. 19 for α_k . In Fig. 18, the estimations aligned well with the ground truth for ECMs indexed 1, 2, and 3, indicating low impact of moderately increased inductance and decreased noise level on RQ element parameter estimation. For ECM4

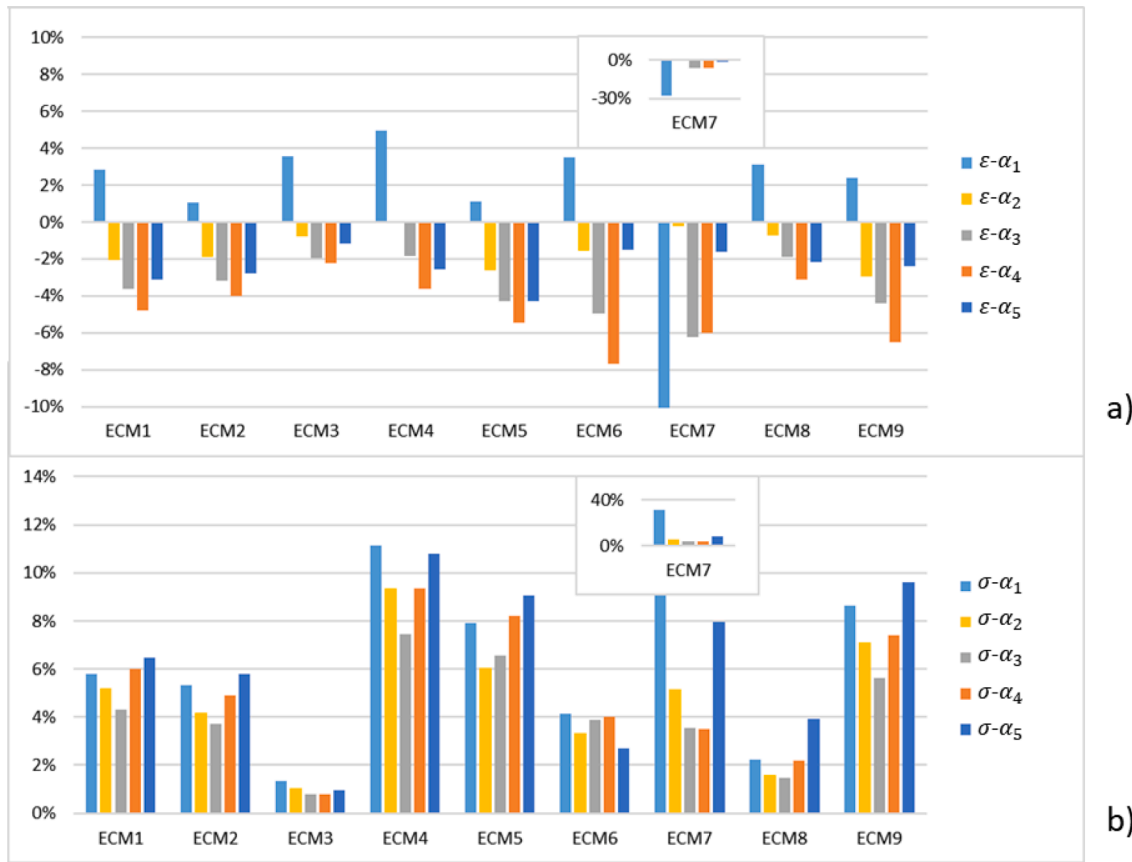


Fig. 19. Estimation errors of α_k of the 5RQ ECMs. (a) average biases, (b) standard deviations.

Table 5

Summary of impacts to ECM parameter estimation of the numerical experiments using the 5RQ ground truth ECMs.

ECM Index	Feature	R_{inf}	L, L'	R_k, τ_k	α_k
1	Comparison	-	-	-	-
2	Greater L	$ e \downarrow, \sigma\uparrow$	$ e \downarrow, \sigma\downarrow$	Not obvious	Not obvious
3	Less noise	$ e \uparrow, \sigma\downarrow$	$ e \uparrow, \sigma\downarrow$	Not obvious	$\sigma\downarrow$
4	Smaller α_k	$ e \uparrow, \sigma\uparrow$	$ e \uparrow, \sigma\uparrow$	$ e \uparrow, \sigma\uparrow$	$\epsilon\uparrow, \sigma\uparrow$
5	Greater R_{inf}	$ e \downarrow, \sigma\downarrow$	$ e \uparrow, \sigma\uparrow$	$ e \uparrow, \sigma\uparrow$ for $(RQ)_5$	$\sigma\uparrow$
6	Greater R_1	Not obvious	Not obvious	Not obvious	$ e \uparrow, \sigma\downarrow$
7	τ_4 closer to τ_3	Not obvious	Not obvious	$ e \uparrow\uparrow, \sigma\uparrow\uparrow$	$ e \uparrow\uparrow, \sigma\uparrow\uparrow$
8	Lower ω_{max}	$ e \uparrow\uparrow, \sigma\uparrow\uparrow$	$ e \uparrow\uparrow, \sigma\uparrow\uparrow$	$ e \uparrow, \sigma\uparrow$ for $(RQ)_5$	$ e \downarrow, \sigma\downarrow$
9	Higher ω_{min}	Not obvious	Not obvious	$ e \uparrow, \sigma\uparrow$ for $(RQ)_1$	$ e \uparrow, \sigma\uparrow$

with lower α_k ground truth values, increased number of misestimated cases appeared. For ECM5 with greater R_{inf} , the estimation error of R_5 and τ_5 increased notably. For ECM6 with greater R_5 , the estimation error of R_k and τ_k remained largely unchanged relative to ECM1, except a minority of samples with misestimated R_k and τ_k for $k = 4, 5$.

For ECM7 with τ_4 closer to τ_3 , the estimation of R_k and τ_k for $k = 1, 2, 5$ in Table 3 remained accurate, yet R_k and τ_k for $k = 3, 4$ was not properly estimated. $(RQ)_3$ and $(RQ)_4$ were misrepresented by two RQ elements with relaxation time close to the geometric average of τ_3 and τ_4 . The DC resistance of one RQ element was close to the sum of R_3 and R_4 , while the DC resistance of the other was close to zero. This

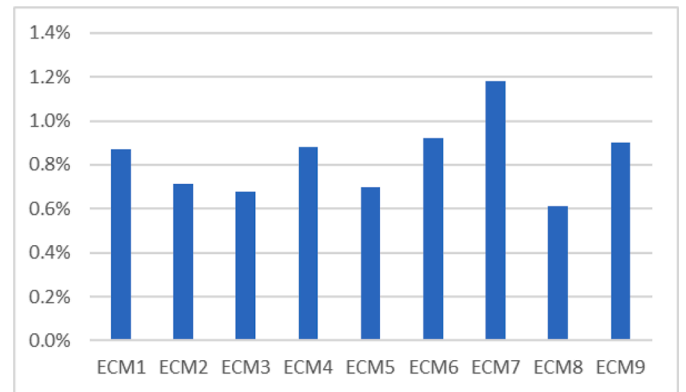


Fig. 20. Averaged ϵ_{ECM} of the 5RQ ECMs.

misallocation of DC resistance can happen when two relaxation processes were not separated as 2 peaks in the DRT result. Such mistake could possibly be solved by manually indicating the initial values of ECM parameters.

ECM8 and ECM9 features a truncated decade of frequency on the high-end and on the low-end, respectively. The estimation error of “fast” RQ elements, or equivalently speaking, RQ elements with short relaxation time, was sensitive to loss of high-frequency impedance data, while the estimation error of “slow” RQ elements was sensitive to loss of low-frequency impedance data.

In Fig. 19a, except ECM7 where all α_k were underestimated, all other ECMs display similar patterns of average biases:..., α_3, α_4 , and α_5 were underestimated, while α_1 was overestimated. The average bias of α_k was mostly less than 10%. In Fig. 19b, the standard deviation of α_k in ECM3

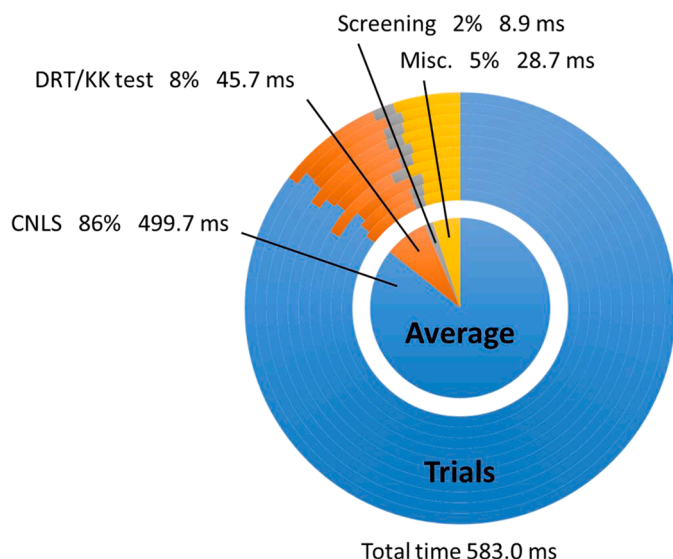


Fig. 21. Time consumption of each part and their proportions fitting ECMs to ECM1 in section 0.

was notably smaller than in ECM1, implying that noise contributes to the estimation uncertainty of α_k . In ECM7, the standard deviation of α_1 was exceptionally high. This can be related to the highly uncertain parameters of $(RQ)_1$, with the estimated R_1 that was close to zero and the estimated τ_1 that spread all across the range of τ_k . Compare $\sigma\alpha_5$ in ECM6 and ECM1, it was clear that relatively greater DC resistance than other RQ elements helps improve the estimation accuracy of α_5 . Consider both

Fig. 19a and b, in ECM8 and ECM9, the loss of high-frequency impedance improved the estimation accuracy of α_k , while the loss of low-frequency impedance undermined the estimation accuracy of α_k .

To sum up, using the proposed algorithm, the impacts of ground truth ECM parameters, noise level, and frequency range of the impedance spectrum were listed in Table 5.

To intuitively demonstrate the ill-posedness of the ECM fitting problem, the relative residual ε_{ECM} of the impedance spectra of each fitted ECM was calculated with Eqs. (17) through (19). The averaged ε_{ECM} over $N_{\text{sim}} = 10000$ samples for all ECMs were plotted in Fig. 20. Despite the vast differences in fitted ECM parameters, the relative residual of impedance spectra was no more than 1.2%, even for the worst case of ECM7. This finding highlights the necessity of using DRT in ECM fitting.

$$Z_{\text{ECM}}(\omega) = R_{\text{inf,fit}} + j\omega L_{\text{fit}} + \omega L'_{\text{fit}} + \sum_k \frac{R_{k,\text{fit}}}{1 + (j\omega\tau_{k,\text{fit}})^{\alpha_{k,\text{fit}}}} \quad (17)$$

$$\varepsilon_{\text{ECM}}(\omega_j) = |Z_{\text{GT}}(\omega_j) - Z_{\text{ECM}}(\omega_j)| \quad (18)$$

$$\varepsilon_{\text{ECM}} = \sqrt{\frac{\sum_{j=1}^N (w(\omega_j) \cdot \varepsilon_{\text{ECM}}(\omega_j))^2}{\sum_{j=1}^N (w(\omega_j))^2}} \quad (19)$$

4.4. Time consumption

The speed of the proposed algorithm was demonstrated through fitting ECMs to samples of ECM1 in the 5RQ ECMs in section 0. Each trial was tested with an impedance spectrum based on identical ground truth ECM parameters and different samples of noise of constant variance. The running time was divided to 4 major parts: CNLS fitting, DRT or KK test,

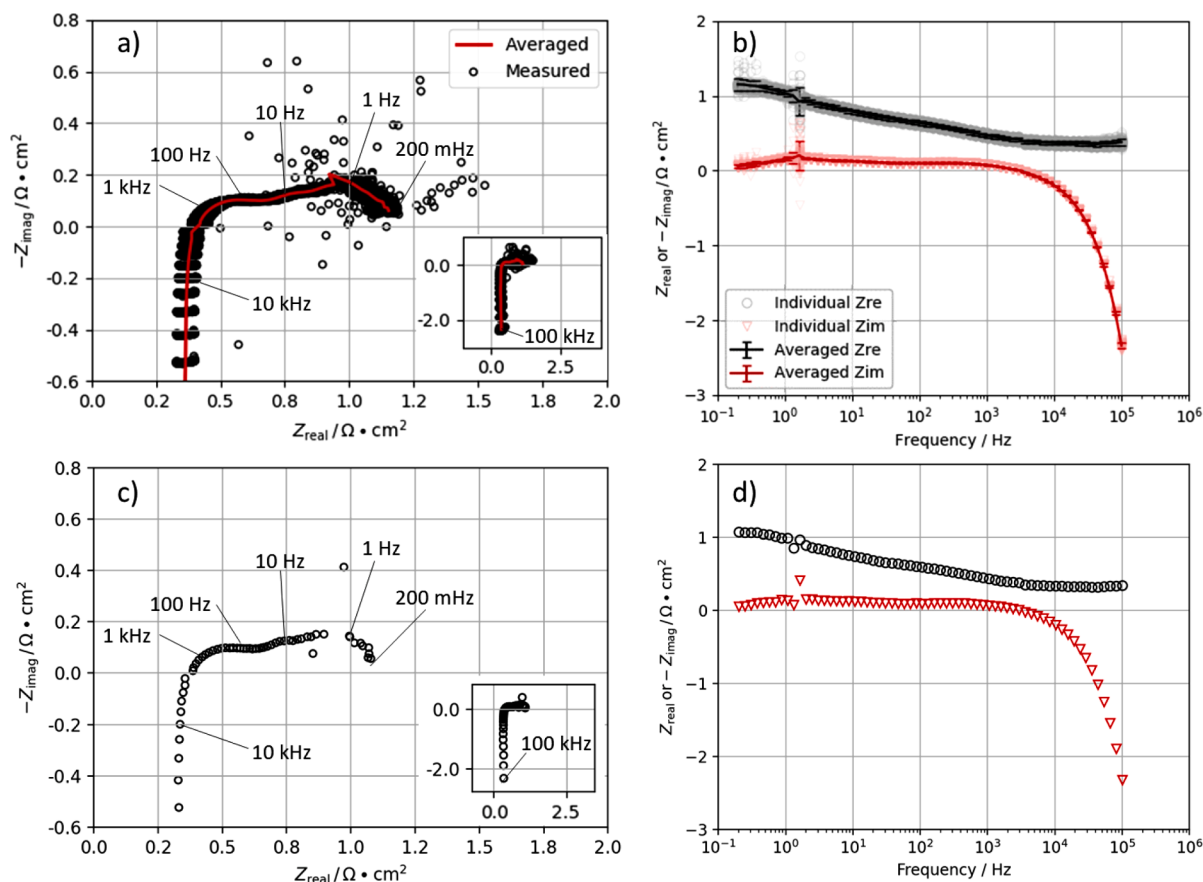


Fig. 22. Measured impedance spectra. (a) Nyquist plot of the measured impedance spectra and their average. (b) The average and standard deviation of the measured impedance. (c) Nyquist plot of a sample of the impedance spectra. (d) The real and imaginary parts of the sample impedance spectrum.

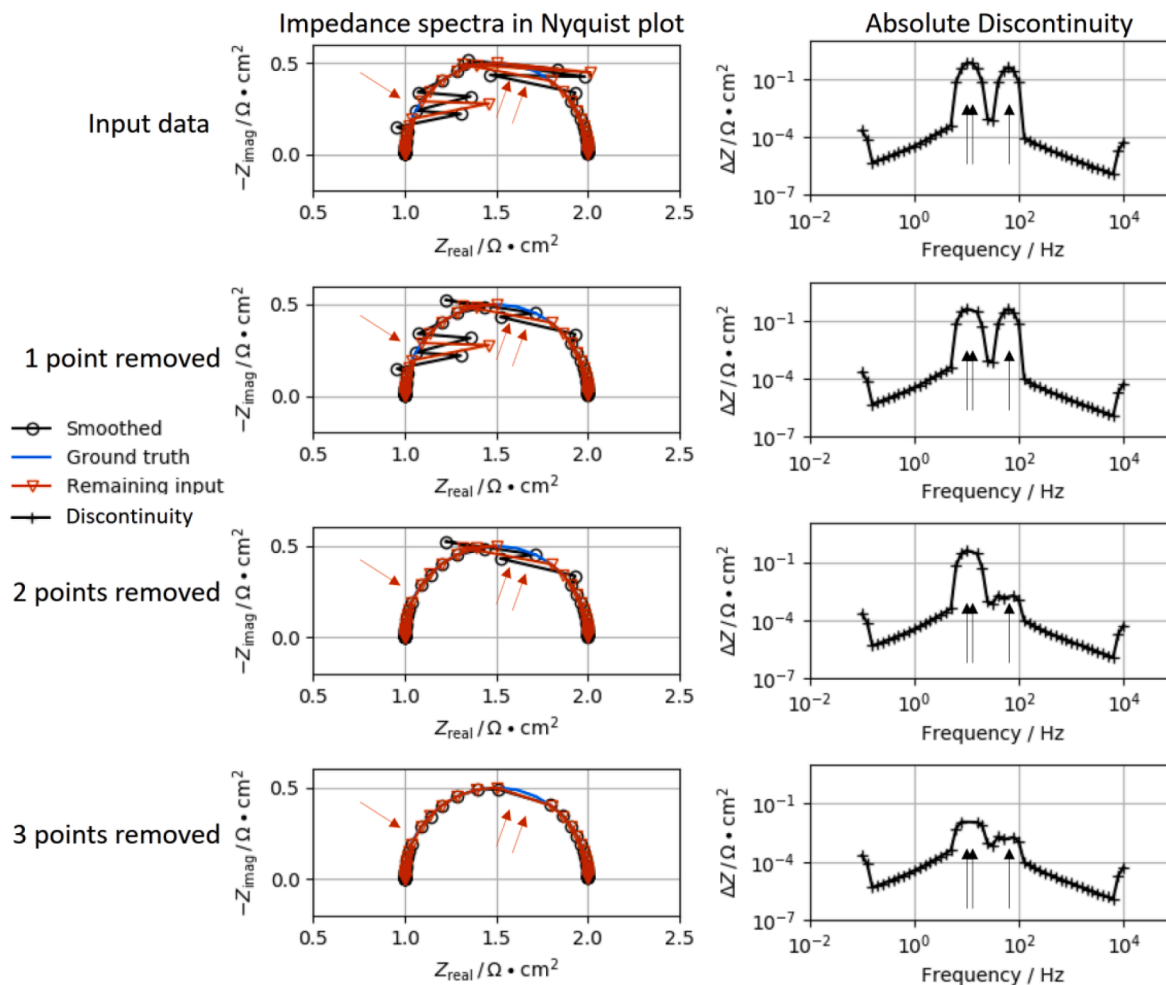


Fig. 23. Data screening process of a simulated impedance spectrum of an RC circuit with 3 wild points. The positions of the wild points are pointed out by the arrows.

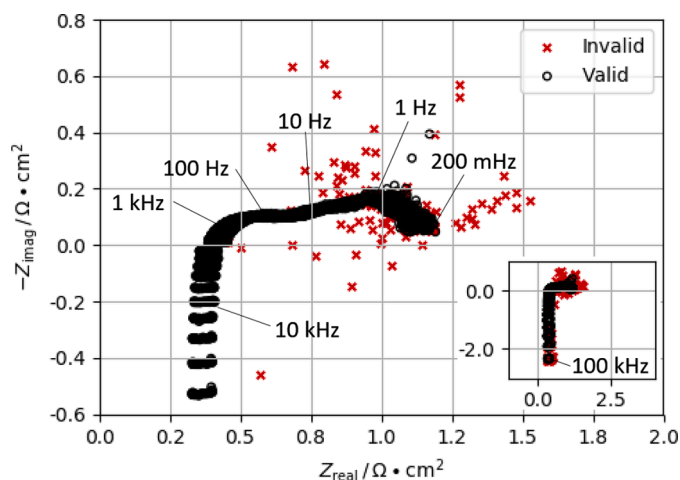


Fig. 24. Nyquist plot of the measured impedance spectra after data screening.

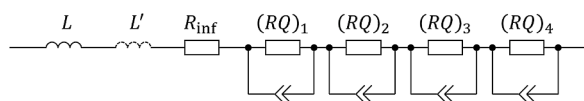


Fig. 25. An example of ECM with $k \in \{1, 2, 3, 4\}$

data screening, and miscellaneous operations such as calculating $\varepsilon_{KK,1}$ to decide whether or not to perform data screening, and calculating the initial guess of ECM parameters, as described in the flow diagram in Fig. 1.

The test was performed on a Surface Pro 5 laptop with an Intel Core i5-7300U CPU while connected to power supply. In total, 10 trials were performed and the running time of each of the 4 major parts was gauged with the Python package “CProfile” for Python 3.6.3. The results were visualized in Fig. 21.

In the 10 trials, the average total time consumption fitting an ECM was 583.0 ms. The CNLS fitting consumed up to 86% of the total runtime, while its preceding part DRT and KK tests consumed only 8% of the total runtime. Data screening took only 2% of the total runtime, yet drastically improved the robustness of DRT, as was demonstrated in Figs. 5 and 6 in section 0.

5. Conclusion and perspectives

The Equivalent Circuit Model (ECM) is a powerful technique to quantitatively analyze and compare Electrochemical Impedance Spectroscopy (EIS) data. There are algorithms that work well on noisy data, yet in many cases, either run time or robustness remains a problem. In this paper, a robust and fast algorithm has been introduced. For ECMs comprising series connected resistor, inductor, and RQ elements, the new algorithm effectively prevented the CNLS fitting from terminating on local minima, enabling fully automatic ECM fitting, by incorporating DRT for initial guesses of ECM parameters. Data screening and weighing subroutines has effectively improved the robustness of DRT against

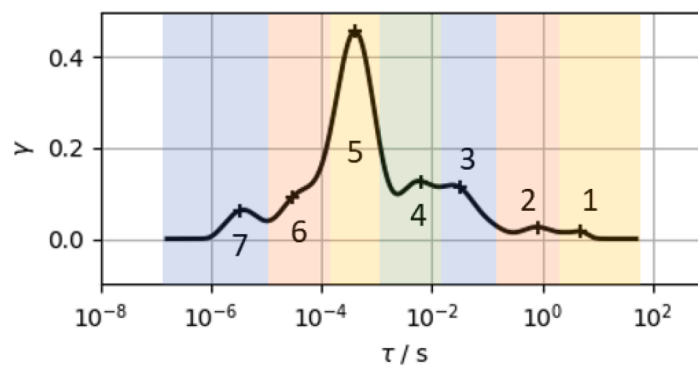


Fig. 26. DRT plot $\gamma(\tau)$ of an aged button solid oxide fuel cell. The peaks are marked 1 through 7. Note that peak 6 is not a local maximal but still recognized. The colored boxes correspond to the integration limits of R_k estimation.

“wild” points. For both experimentally measured and simulated noisy EIS data, the new algorithm reduced the impact of noise drastically. The time consumption fitting an ECM has been shortened to within 1 s.

The central ideas in this algorithm are:

- Identifying mathematically ill-posed or highly non-linear problems in ECM fitting, including DRT evaluation and CNLS fitting of ECM.
- Data screening and weight assigning prior to DRT and ECM analysis, using input data smoothness and KK test residuals.
- Reducing the impact of noise to the ill-posed DRT problem by properly weighing the input data.
- Increasing the chance of finding the global optima in the CNLS fitting of ECM with proper initial guesses extracted from DRT.

In future studies on ECM fitting, there are yet a few problems to be resolved:

- The applicability of DRT-assisted ECM fitting algorithms to ECM structures other than resistor + inductor + multiple RQ elements.
- Uncertainty estimation of DRT and ECM parameters, including their upper and lower limits given certain confidence probability.
- The compatibility with RQ elements with negative DC resistance of DRT-assisted ECM fitting algorithms.

Appendix: detailed methodology

Weighted KK test

The Kramers-Kronig test (KK test) is a validation technique of impedance spectra, derived from the correlation between the real part and the imaginary part of the impedance of causal systems [36], namely the Kramers-Kronig relation. Mathematically, the KK relation at positive angular frequencies [37] is written as Eqs. (20) and (21), $Z_{\text{real}}(\omega)$ and $Z_{\text{imag}}(\omega)$ are real-valued functions corresponding to the real part and the imaginary part of the impedance $Z(\omega) = Z_{\text{real}}(\omega) + jZ_{\text{imag}}(\omega)$.

$$Z_{\text{real}}(\omega) = \frac{2}{\pi} \int_0^{+\infty} \frac{\omega' \cdot Z_{\text{imag}}(\omega')}{\omega^2 - \omega'^2} d\omega' \quad (20)$$

$$Z_{\text{imag}}(\omega) = -\frac{2}{\pi} \int_0^{+\infty} \frac{\omega \cdot Z_{\text{real}}(\omega')}{\omega^2 - \omega'^2} d\omega' \quad (21)$$

As pointed out by Schönleber et al. [37], evaluating Eqs. (20) and (21) for KK test suffers from missing frequencies as $\omega \rightarrow 0$ and $\omega \rightarrow +\infty$. Therefore, similar to Schönleber et al. [37], the KK test in this study is equivalently fitting an ECM to the measured impedance. The ECM comprises the following elements in series connection: N RC elements (each comprising a resistor R_k in parallel with a capacitor C_k), a series resistor R_{inf} , an inductor L , a pseudo inductor L' accounting for the inductance-like phenomenon due to skin effects in the conductors or finite common-mode rejection ratio of the measuring instrument or other similar factors. N equals the number of frequencies in the input impedance spectrum. Fitting the ECM is equivalently minimizing Loss in Eq. (22), $w(\omega_j)$ is the weight at the j th frequency point. Loss_j is expressed in Eqs. (23), $\tau_k = R_k C_k$ is the time constant of the k th RC element.

For demonstration, the algorithm has been implemented in Python with Graphical User Interface (GUI), as part of the software “EISART”. The software is freely available at github.com/leehangyue/EISART.

CRediT authorship contribution statement

Hangyue Li: Conceptualization, Methodology, Software, Investigation, Writing – original draft. **Zewei Lyu:** Investigation, Validation, Writing – review & editing. **Minfang Han:** Supervision, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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$$Loss = \sum_j |w(\omega_j) \cdot Loss_j|^2 \quad (22)$$

$$Loss_j = \sum_{k=1}^N \frac{R_k}{1 + j\omega_j \tau_k} + R_{inf} + j\omega_j L + \omega_j L' - Z(\omega_j) \quad (23)$$

With the definition of the parameter vector $\mathbf{x}$ of the ECM in Eq. (24) and the transformation matrix $\mathbf{A}$ in Eq. (25), $Loss$ is written in a more compact form in Eq. (26), the superscript T denotes the transpose, the superscript H denotes the conjugate transpose. To linearize the ECM fitting problem, τ_k is fixed and its value is given in Eq. (27), note that N equals the number of frequencies in the input impedance spectrum. $\mathbf{W} = \text{diag}(\mathbf{w})$ is the diagonal matrix of the weights $\mathbf{w}$.

$$\mathbf{x} = [R_{inf} \quad L \quad L' \quad R_1 \quad \cdots \quad R_N]^T \quad (24)$$

$$\mathbf{A} = \begin{bmatrix} 1 & j\omega_1 & \omega_1 & \frac{1}{1+j\omega_1\tau_1} & \cdots & \frac{1}{1+j\omega_1\tau_N} \\ 1 & j\omega_2 & \omega_2 & \frac{1}{1+j\omega_2\tau_1} & \cdots & \frac{1}{1+j\omega_2\tau_N} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & j\omega_N & \omega_N & \frac{1}{1+j\omega_N\tau_1} & \cdots & \frac{1}{1+j\omega_N\tau_N} \end{bmatrix} \quad (25)$$

$$Loss = (\mathbf{Ax} - \mathbf{Z})^H \mathbf{W}^T \mathbf{W} (\mathbf{Ax} - \mathbf{Z}) \quad (26)$$

$$\tau_k = \frac{1}{\omega_1} \left(\frac{\omega_1}{\omega_N} \right)^{\frac{k-1}{N-1}} \quad (27)$$

$$\mathbf{W} = \begin{bmatrix} w(\omega_1) & 0 & \cdots & 0 \\ 0 & w(\omega_2) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & w(\omega_N) \end{bmatrix} \quad (28)$$

To minimize $Loss$, Eq. (26) is rearranged as Eq. (29), which is a typical quadratic programming problem with elementwise constrain $x_m \geq 0$, $m \in \{1, 4, 5, \dots, N+3\}$ because $R_{inf} \geq 0$ and $R_k \geq 0$, $k \in \{1, 2, \dots, N\}$. $\mathbf{A}_{real}$ and $\mathbf{A}_{imag}$ are the real and imaginary part of the complex-valued matrix $\mathbf{A}$, $\mathbf{Z}_{real}$ and $\mathbf{Z}_{imag}$ are the real and imaginary part of the input impedance. In this study, the Python package “quadprog” implementing the algorithm by Goldfarb and Idnani [38] was employed to solve for $\mathbf{x}$.

$$Loss = \mathbf{x}^T \left(\mathbf{A}_{real}^T \mathbf{W}^T \mathbf{W} \mathbf{A}_{real} + \mathbf{A}_{imag}^T \mathbf{W}^T \mathbf{W} \mathbf{A}_{imag} \right) \mathbf{x} - \left(\mathbf{A}_{real}^T \mathbf{W}^T \mathbf{W} \mathbf{Z}_{real} + \mathbf{A}_{imag}^T \mathbf{W}^T \mathbf{W} \mathbf{Z}_{imag} \right) \mathbf{x} + \left(\mathbf{Z}_{real}^T \mathbf{W}^T \mathbf{W} \mathbf{Z}_{real} + \mathbf{Z}_{imag}^T \mathbf{W}^T \mathbf{W} \mathbf{Z}_{imag} \right) \quad (29)$$

Suppose $\mathbf{x}^*$ is the solution that minimizes $Loss$. Then, the estimated impedance spectrum $\mathbf{Z}_{KK}$ is given by Eq. (30).

$$\mathbf{Z}_{KK} = \mathbf{Ax}^* \quad (30)$$

Impedance data screening

In an impedance spectrum, noise is the deviation of impedance from ground truth due to unconsidered processes or unexpected incidents. In this study, noise was modeled as Gaussian white noise.

Fig. 22 shows the average and standard deviation of measured impedance $Z(\omega)$ at each frequency. Observe Fig. 22b, the impedance at 1.6 Hz ($Z(1.6 \text{ Hz})$) exhibited considerably higher standard deviation than other frequencies. In a single impedance spectrum, chances are high that $Z(1.6 \text{ Hz})$ exhibited the highest discontinuity with its neighboring frequencies. To quantify discontinuity, a second-order Savitzky-Golay filter is applied for a smoothed impedance $Z_{smooth}(\omega)$ at each frequency. The impedance at each frequency is estimated from N_{SG} other data points at nearest frequencies using a second-order polynomial. The coefficients of the polynomial are determined through least squares regression to the N_{SG} data pairs of the form $Z(\omega) \cdot \ln(\omega)$, as specified in Eqs. (31) through (34). The discontinuity is thereafter defined as $\varepsilon(\omega) = |Z(\omega) - Z_{smooth}(\omega)|$. In this study, $N_{SG} = 4$ is chosen. The calculation process of $Z_{smooth, re}(\omega)$ is demonstrated in Eqs. (31) through (34). $Z_{smooth, im}(\omega)$ is calculated similarly.

$$\mathbf{X}^T \mathbf{X} \mathbf{a}_{re} = \mathbf{X}^T \mathbf{Z}_{re} \quad (31)$$

$$\mathbf{X} = \begin{bmatrix} 1 & \ln(\omega_{j1}) & \ln^2(\omega_{j1}) \\ 1 & \ln(\omega_{j2}) & \ln^2(\omega_{j2}) \\ 1 & \ln(\omega_{j3}) & \ln^2(\omega_{j3}) \\ 1 & \ln(\omega_{j4}) & \ln^2(\omega_{j4}) \end{bmatrix} \quad (32)$$

$$\mathbf{a}_{re} = [a_{re,1} \quad a_{re,2} \quad a_{re,3}]^T \quad (33)$$

$$Z_{smooth, re}(\omega_j) = a_{re,1} + a_{re,1} \ln(\omega_j) + a_{re,2} \ln^2(\omega_j) \quad (34)$$

As is shown in the first row “Input data” in Fig. 23, there is high discontinuity at and around a wild point. Therefore, distinguishing a wild point

from its surrounding “normal” data points by discontinuity is challenging. To avoid “normal” data points being screened out, only one data point with the greatest discontinuity is removed at a time. After removing a data point, the discontinuity of its nearby points are recalculated excluding the removed point. The overall discontinuity $\varepsilon_{it} = \text{average}(\varepsilon(\omega_j))_{j \in S_{it}}$ is evaluated in the end of each iteration, it is the iteration count, S_{it} is the set of remaining indices of j at the end of the it th iteration. The screening continues until $r_{\varepsilon, it} = \frac{\varepsilon_{it}}{\varepsilon_{it-1}} < r_{\varepsilon}^* = 0.5$ and $it \geq it_{\min} = 5$. To retain enough data for analysis, the iteration terminates when it reaches $it_{\max} = 15$, the maximum number of iterations. The flow diagram of data screening is shown in the box “Data Screening” in Fig. 1.

The whole data screening process is shown in Fig. 23. In a simulated impedance spectrum of an RC circuit, 3 wild points were deliberately added. After 3 iterations removing data points of high discontinuity, the 3 artificial wild points were successfully removed, see the bottom row “3 points removed” in Fig. 23.

This data screening algorithm was applied to the measured impedance spectra that are mentioned in 0.3.1 Noise Measurement. The results are plotted in Fig. 24. Note that most wild points were screened out by the algorithm. The data points marked invalid are assigned zero weights and passed on to the following steps.

Weight adjusting

The flow diagram of weight adjusting and DRT is shown in the box named “Weight Adjusting” in Fig. 1. The first step “Adjust $w(\omega_j)$ ” will be explained later in this section. After adjusting $w(\omega_j)$, the following step “KK Test” yields an estimated impedance spectrum $Z_{KK}(\omega_j)$, together with the residuals $\varepsilon(\omega_j)$ in Eq. (35) and its weighed RMS ε , as given by Eq. (36). ε is compared to the threshold $\varepsilon_{KK,2}^*$. If $\varepsilon \leq \varepsilon_{KK,2}^*$ and the iteration count it is greater than $it_{\max}$, then exit weight adjusting. Otherwise, start another iteration. In this study, $\varepsilon_{KK,2}^* = 0.5\%$. In this study, $it_{\max} = 1$.

$$\varepsilon(\omega_j) = |Z(\omega_j) - Z_{KK}(\omega_j)| \quad (35)$$

The first step “Adjust $w(\omega_j)$ ” includes the 2 following subroutines: “calculate weight suggestions” and “update weights”. In the first subroutine, the weight adjusting suggestions $w_s(\omega_j)$ are calculated from the relative “prominence” of KK test residual, to capture the data points that deviate from the rest but still weighed high, see Eq. (37). The weight adjusting suggestions $w_s(\omega_j)$ are high-pass filtered by convolving with kernel $h = [-0.25 \ 0.5 \ -0.25]^T$, to prevent deviated data points corrupting the KK test locally and hence the weight of neighboring “normal” data points being mistakenly reduced.

$$\varepsilon = \sqrt{\frac{\sum_{j=1}^N (w(\omega_j) \cdot \varepsilon(\omega_j))^2}{\sum_{j=1}^N (w(\omega_j))^2}} \quad (36)$$

$$w_s(\omega_j) = \sum_{q=-1}^1 h_q \left(\frac{\varepsilon}{\varepsilon + w(\omega_{j-q}) \cdot \varepsilon(\omega_{j-q})} \right) \quad (37)$$

$$w(\omega_j) = 1, j(1 \text{ or } j)N$$

Instead of updating the weights with the weight suggestions, the inversion of weights are updated instead across iterations, to avoid the weights of the identified wild points being recovered too quickly, see Eq. (38), $w_{it}(\omega_j)$ is the weight at the j th frequency in the it th iteration, $\zeta = 0.5$ is the weight update ratio, $w_{\min} = 10^{-6}$ is a small number to avoid zero division.

$$w_{it}(\omega_j)^{-1} = \zeta \cdot (w_s(\omega_j))^{-2} + (1 - \zeta) \cdot (\max(w_{\min}, w_{it-1}(\omega_j)))^{-1} \quad (38)$$

Modified hyper- λ DRT

Based on the work of Effat and Ciucci [11], the DRT analysis is converted to a minimization problem that seeks minimum of Loss in Eq. (39). B_{real} and B_{imag} are the real and the imaginary parts of the transformation matrix B in Eq. (40). The elements $b_{\omega_j \tau_i}$ are given by Eq. (41), $\phi(x)$ is the radial basis function. In this study, $\phi(x)$ is chosen as Eq. (42), $\sigma = \frac{2}{N-1} \left| \ln \frac{\omega_1}{\omega_N} \right|$. The subsequent distribution function of relaxation time $\gamma(\tau)$ is given by Eq. (43), where x' is the last N elements of x , excluding R_{inf} , L and L' . Similar to Eq. (28), Λ is the diagonal matrix of the regularization factors of the elements of x , as shown in Eq. (44).

$$\text{Loss} = x^T \left(B_{\text{real}}^T W^T W B_{\text{real}} + B_{\text{imag}}^T W^T W B_{\text{imag}} \right) x - \left(B_{\text{real}}^T W^T W Z_{\text{real}} + B_{\text{imag}}^T W^T W Z_{\text{imag}} \right) x + x^T \Lambda M \Lambda x \quad (39)$$

$$B = \begin{bmatrix} 1 & j\omega_1 & \omega_1 & b_{\omega_1 \tau_1} & \cdots & b_{\omega_1 \tau_N} \\ 1 & j\omega_2 & \omega_2 & b_{\omega_2 \tau_1} & \cdots & b_{\omega_2 \tau_N} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & j\omega_N & \omega_N & b_{\omega_N \tau_1} & \cdots & b_{\omega_N \tau_N} \end{bmatrix} \quad (40)$$

$$b_{\omega_j \tau_i} = \int_{-\infty}^{+\infty} \frac{\phi\left(\ln\left(\frac{\tau}{\tau_i}\right)\right) d\ln\tau}{1 + j\omega_j \tau} \quad (41)$$

$$\phi(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{x^2}{2\sigma^2}\right) \quad (42)$$

$$\gamma(\tau) = \sum_{j=1}^N x'_j \cdot \phi\left(\ln\left(\frac{\tau}{\tau_j}\right)\right) \quad (43)$$

$$\Lambda = \text{diag}\left(\begin{bmatrix} 0^T & \lambda^T \end{bmatrix}^T\right) = \text{diag}([0 \quad 0 \quad 0 \quad \lambda_1 \quad \dots \quad \lambda_N]^T) \quad (44)$$

To incorporate the weights from the previous steps instead of automatically determine the weights, an approach different from the work of Effat and Ciucci [11] was chosen. The weights from the previous step are used without modification, and the regularization factors are calculated from the weights. Considering that the impedance at each frequency is related to processes at multiple relaxation time, as $b_{\omega\tau_i} \neq 0$, $i \neq j$, the regularization factors are calculated from “smoothed” weights, see Eqs. (45) and (46). $\psi(x) = \exp\left(-\frac{x^2}{2\sigma^2}\right)$ is the spread kernel with spreading radius $\sigma = \ln 10$, equivalently one decade in relaxation time. Eq. (46) was designed to ensure:

- $\lambda_0 \leq \lambda_i \leq 1$
- λ_i is larger at relaxation time corresponding to low weights. Equivalently speaking, stronger regularization is applied to frequencies with lower weights.

In this study, $\lambda_0 = 10^{-3}$ was chosen.

$$w_{\text{spread}}(\tau) = \sum_{j=1}^N \psi(\ln(\omega_j \tau)) \cdot (1 - w(\omega_j)) \quad (45)$$

$$\lambda_i = \min\left(1, \frac{\lambda_0}{\left(\frac{w_{\text{spread}}(\tau_i)}{\max_i w_{\text{spread}}(\tau_i)}\right)^{n_i}}\right) \quad (46)$$

Similar to weighed KK tests, the Python package “quadprog” implementing the algorithm by Goldfarb and Idnani [38] is employed to minimize Loss and solve for x . Except the 2 elements corresponding to L , and L' , all other elements of x are subject to restriction $x_i \geq 0$. L and L' are unbounded.

Note that for impedance spectra with sufficient frequency resolution typically no lower than 10 points per decade, σ in Eq. (42) may adopt sufficiently small values that $b_{\omega\tau_i} \approx 1/(1 + j\omega\tau_i)$. Due to this mathematical relation, a relaxation process at a certain relaxation time mainly affects the trend of impedance with frequency around the characteristic frequency of the relaxation time $f_i = 1/(2\pi\tau_i)$. For impedance spectra that are noisier at certain frequencies, the uncertainty of DRT is higher around the relaxation time corresponding to the noisy frequency band. The locality of noise-related uncertainty in DRT serves as the starting point of data quality-based weighting in DRT evaluation.

DRT-assisted ECM fitting

$$Z_{\text{ECM}}(\omega) = R_{\text{inf}} + j\omega L + \omega L' + \sum_k \frac{R_k}{1 + (j\omega\tau_k)^{\alpha_k}} \quad (47)$$

In this study, the final output ECM is intended to reflect physiochemical processes in the test object. It comprises the following elements in series connection: a series resistor R_{inf} , an inductor L , a pseudo inductor L' accounting for the inductance-like phenomenon due to skin effects in the conductors or finite common-mode rejection ratio of the measuring instrument or other similar factors, and n RQ elements (each comprising a resistor R_k in parallel with a constant phase element Q_k) that are intended to represent the charge transfer or species transport processes in the test object. Fig. 25 shows a typical ECM with 4 RQ elements. Each RQ element is characterized with its DC resistance R_k , characteristic time constant τ_k , and spread exponent α_k of relaxation time distribution. The impedance spectrum of the ECM is given by Eq. (47).

As shown in the box named “DRT-assisted ECM Fitting” in Fig. 1, an initial guess of ECM parameters is determined through 3 subroutines: determine the number N_{RQ} of RQ elements in the ECM along with their relaxation time τ_k , estimate R_k , and initialize α_k , $k \in \{1, 2, \dots, N_{\text{RQ}}\}$. For an intuitive demonstration of the effects of changing the parameters of an RQ element, please see Chapter 2 of the doctoral thesis of Leonide [39]. The series resistance R_{inf} , the inductance L , and the pseudo inductance L' are determined by DRT analysis in the previous step.

Intuitively speaking, the number of RQ elements N_{RQ} corresponds to the number peaks in the DRT plot, as shown in Fig. 26. Besides the peaks in the DRT plot, there can be a “shoulder” corresponding to a small peak next to a big one. Marking barely the peaks in DRT plot may not suffice to identify all the potential relaxation processes. Therefore, $\gamma_{\text{d2}}(\tau)$, the smoothed second-order derivative of γ to $\ln(\tau)$ is evaluated by convolving γ with $s(x)$, as shown in Eqs. (48) and (49). In this study, $\sigma_{\text{peak}} = 0.1\ln(10)$ was chosen as a rule of thumb to balance sensitivity to wide and narrow peaks. Next, $\gamma_{\text{d2}}(\tau)$ and $\gamma(\tau)$ are combined into $p(\tau)$, see Eq. (50), so that the peak prominence of $p(\tau)$ roughly match the DC resistances of each potential relaxation process.

$$s(x) = \frac{d^2}{dx^2} \left(\frac{1}{\sqrt{2\pi}\sigma_{\text{peak}}} \exp\left(-\frac{x^2}{2\sigma_{\text{peak}}^2}\right) \right) \quad (48)$$

$$\gamma_{\text{d2}}(\tau) = \int_{-\infty}^{+\infty} \gamma(\tau \cdot \exp(-x)) \cdot s(x) dx \quad (49)$$

$$p(\tau) = 0.81(1 - (1 - \gamma_{\text{d2}}(\tau))^3) + 0.09\gamma_{\text{d2}}(\tau) + 0.1\gamma(\tau) \quad (50)$$

To extract N_{RQ} and τ_k from $p(\tau)$, the function “find_peaks” in Python package “scipy.signal” was employed. The peaks with prominences greater than 1% of the maximum prominence are marked. N_{RQ} is therefore the number of marked peaks and τ_k are the corresponding peak positions. If N_{RQ} is

known, then the N_{RQ} peaks with maximum prominences are marked.

With τ_k specified, R_k are estimated by integrating $\gamma(\tau)$ around each τ_k . The integration limits are simply set to the geometric average of each adjacent pair of τ_k . For τ_k at the edge, the integration limit at the outer side is set to the edge of the range of τ .

To estimated α_k , the analytical DRT of an RQ element in Eq. (51) is considered, according to the mathematical deduction by Boukamp [40]. Inserting $\tau = \tau_k$, the peak height of an RQ element is obtained in Eqs. (42) and (52). Arrange Eq. (52), the analytical expression of α_k is obtained in Eq. (53). Note that the peak height of radial basis function-based DRT is limited as α_k approaches 1, Eq. (53) is corrected as Eq. (54), with the correction factor $\xi = 0.9$. To estimated α_k from numerically evaluated DRT, $\gamma_{\text{GT,peak}}$ is approximated with the peak value $\gamma_{\text{peak},k}$ of $\gamma(\tau)$ at the corresponding τ_k . Note that in the presence of overlapping peaks, $\gamma_{\text{peak},k}$ can be overestimated, and hence the corresponding α_k can be overestimated.

$$\gamma_{\text{GT}}(\tau) = \sum_k \gamma_{\text{GT},k}(\tau) = \sum_k \frac{R_k}{2\pi} \frac{\sin(\alpha_k \pi)}{\cosh(\alpha_k \ln(\frac{\tau_k}{\tau})) + \cos(\alpha_k \pi)} \quad (51)$$

$$\gamma_{\text{GT,peak}} = \gamma_{\text{GT},k}(\tau_k) = \frac{R_k}{2\pi} \frac{\sin(\alpha_k \pi)}{1 + \cos(\alpha_k \pi)} = \frac{R_k}{2\pi} \tan\left(\frac{\alpha_k \pi}{2}\right) \quad (52)$$

$$\alpha_k = \frac{2}{\pi} \arctan\left(\frac{2\pi \gamma_{\text{GT,peak}}}{R_k}\right) \quad (53)$$

$$\alpha_k = \frac{1}{\xi} \frac{2}{\pi} \arctan\left(\xi \frac{2\pi \gamma_{\text{GT,peak}}}{R_k}\right) \quad (54)$$

With reasonable initial values of R_k , τ_k , and α_k , the CNLS fit refines R_k , τ_k , and α_k by minimizing Loss, the weighed sum of squared errors, in Eq. (55).

Note that $R_{\text{pol}} = \sum_{k=1}^n R_k = \int_0^{\infty} \gamma(\tau) d\tau$ is a constant throughout the CNLS fit, R_1 is calculated as $R_1 = R_{\text{pol}} - \sum_{k=2}^n R_k$ and only $R_k, k \geq 2$ are manipulated by the minimizer during the fit.

$$\text{Loss} = \sum_{j=1}^N w(\omega_j) \cdot |Z(\omega_j) - Z_{\text{ECM}}(\omega_j)|^2 \quad (55)$$

The minimization algorithm “Truncated Newton Conjugate” (TNC) implemented in the Python package “scipy.optimize.minimize” was employed to refine ECM parameters subject to constraints: $R_k > 0$, $\tau_k > 0$, and $0.4 \leq \alpha_k \leq 1.0$.

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